

CHAPTER I

General Topology

A round man cannot be expected to fit in a square hole right away. He must have time to modify his shape.

MARK TWAIN

1. Metric Spaces

We are all familiar with the notion of distance in euclidean n -space: If $\mathbf{x}$ and $\mathbf{y}$ are points in $\mathbf{R}^n$ then

$$\text{dist}(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}.$$

This notion of distance permits the definition of continuity of functions from one euclidean space to another by the usual ϵ - δ definition:

$f: \mathbf{R}^n \rightarrow \mathbf{R}^k$ is continuous at $\mathbf{x} \in \mathbf{R}^n$ if, given $\epsilon > 0$,

$$\exists \delta > 0 \ni \text{dist}(\mathbf{x}, \mathbf{y}) < \delta \Rightarrow \text{dist}(f(\mathbf{x}), f(\mathbf{y})) < \epsilon.$$

Although the spaces of most interest to us in this book are subsets of euclidean spaces, it is useful to generalize the notion of “space” to get away from such a hypothesis, because it would be very complicated to try to verify that spaces we construct are always of this type. In topology, the central notion is that of continuity. Thus it would usually suffice for us to treat “spaces” for which we can give a workable definition of continuity.

We could define continuity as above for any “space” which has a suitable notion of distance. Such spaces are called “metric spaces.”

1.1. Definition. A *metric space* is a set X together with a function

$$\text{dist}: X \times X \rightarrow \mathbf{R},$$

called a *metric*, such that the following three laws are satisfied:

- (1) (positivity) $\text{dist}(x, y) \geq 0$ with equality $\Leftrightarrow x = y$;
- (2) (symmetry) $\text{dist}(x, y) = \text{dist}(y, x)$; and
- (3) (triangle inequality) $\text{dist}(x, z) \leq \text{dist}(x, y) + \text{dist}(y, z)$.

In a metric space X we define the “ ϵ -ball,” $\epsilon > 0$, about a point $x \in X$ to be

$$B_\epsilon(x) = \{y \in X \mid \text{dist}(x, y) < \epsilon\}.$$

Also, a subset $U \subset X$ is said to be “open” if, for each point $x \in U$, there is an ϵ -ball about x completely contained in U . A subset is said to be “closed” if its complement is open. If $y \in B_\epsilon(x)$ and if $\delta = \epsilon - \text{dist}(x, y)$ then $B_\delta(y) \subset B_\epsilon(x)$ by the triangle inequality. This shows that all ϵ -balls are open sets.

It turns out that, for metric spaces, continuity can be expressed completely in terms of open sets:

1.2. Proposition. *A function $f: X \rightarrow Y$ between metric spaces is continuous $\Leftrightarrow f^{-1}(U)$ is open in X for each open subset U of Y .*

PROOF. If f is continuous and $U \subset Y$ is open and $f(x) \in U$ then there is an $\epsilon > 0$ such that $B_\epsilon(f(x)) \subset U$. By continuity, there is a $\delta > 0$ such that f maps the δ -ball about x into $B_\epsilon(f(x))$. This means that $B_\delta(x) \subset f^{-1}(U)$. This implies that $f^{-1}(U)$ is open.

Conversely, suppose $f(x) = y$ and that $\epsilon > 0$ is given. By hypothesis, $f^{-1}(B_\epsilon(y))$ is open and contains x . Therefore, by the definition of an open set, there is a $\delta > 0$ such that $B_\delta(x) \subset f^{-1}(B_\epsilon(y))$. It follows that if $\text{dist}(x, x') < \delta$ then $f(x') \in B_\epsilon(y)$, and so $\text{dist}(f(x), f(x')) < \epsilon$, proving continuity in the ϵ - δ sense. $\square$

The only examples of metric spaces we have discussed are euclidean spaces and, of course, subsets of those. Even with those, however, there are other reasonable metrics:

$$\text{dist}_2(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|,$$

$$\text{dist}_3(\mathbf{x}, \mathbf{y}) = \max(|x_i - y_i|).$$

It is not hard to verify, from the following proposition, that these three metrics give the same open sets, and so behave identically with respect to continuity (for maps into or out of them).

1.3. Proposition. *If dist_1 and dist_2 are metrics on the same set X which satisfy the hypothesis that for any point $x \in X$ and $\epsilon > 0$ there is a $\delta > 0$ such that*

$$\text{dist}_1(x, y) < \delta \quad \Rightarrow \quad \text{dist}_2(x, y) < \epsilon,$$

and

$$\text{dist}_2(x, y) < \delta \quad \Rightarrow \quad \text{dist}_1(x, y) < \epsilon,$$

then these metrics define the same open sets in X .

PROOF. The proof is an easy exercise in the definition of open sets and is left to the reader. $\square$

PROBLEMS

1. Consider the set X of all continuous real valued functions on $[0, 1]$. Show that

$$\text{dist}(f, g) = \int_0^1 |f(x) - g(x)| dx$$

defines a metric on X . Is this still the case if continuity is weakened to integrability?

2. ϕ If X is a metric space and x_0 is a given point in X , show that the function $f: X \rightarrow \mathbf{R}$ given by $f(x) = \text{dist}(x, x_0)$ is continuous.
3. ϕ If A is a subset of a metric space X then define a real valued function d on X by $d(x) = \text{dist}(x, A) = \inf\{\text{dist}(x, y) | y \in A\}$. Show that d is continuous. (*Hint*: Use the triangle inequality to show that $|d(x_1) - d(x_2)| \leq \text{dist}(x_1, x_2)$.)

2. Topological Spaces

Although most of the spaces that will interest us in this book are metric spaces, or can be given the structure of metric spaces, we will usually only care about continuity of mappings and not the metrics themselves. Since continuity can be expressed in terms of open sets alone, and since some constructions of spaces of interest to us do not easily yield to construction of metrics on them, it is very useful to discard the idea of metrics and to abstract the basic properties of open sets needed to talk about continuity. This leads us to the notion of a general “topological space.”

2.1. Definition. A *topological space* is a set X together with a collection of subsets of X called “open” sets such that:

- (1) the intersection of two open sets is open;
- (2) the union of any collection of open sets is open; and
- (3) the empty set $\emptyset$ and whole space X are open.

Additionally, a subset $C \subset X$ is called “closed” if its complement $X - C$ is open.

Topological spaces are much more general than metric spaces and the range of difference between them and metric spaces is much wider than that between metric spaces and subspaces of euclidean space. For example, it is possible to talk about convergence of sequences of points in metric spaces with little difference from sequences of real numbers. Continuity of functions can be described in terms of convergence of sequences in metric spaces. One can also talk about convergence of sequences in general topological spaces but that no longer is adequate to describe continuity (as we shall see later). Thus it is necessary to exercise care in developing the theory of general topological spaces. We now begin that development, starting with some further basic definitions.

2.2. Definition. If X and Y are topological spaces and $f: X \rightarrow Y$ is a function, then f is said to be *continuous* if $f^{-1}(U)$ is open for each open set $U \subset Y$. A *map* is a continuous function.

Since closed sets are just the complements of open sets and since inverse images preserve complements (i.e., $f^{-1}(Y - B) = X - f^{-1}(B)$), it follows that a function $f: X \rightarrow Y$ is continuous $\Leftrightarrow f^{-1}(F)$ is closed for each closed set $F \subset Y$.

2.3. Definition. If X is a topological space and $x \in X$ then a set N is called a *neighborhood* of x in X if there is an open set $U \subset N$ with $x \in U$.

Note that a neighborhood is not necessarily an open set, and, even though one usually thinks of a neighborhood as “small,” it need not be: the entire space X is a neighborhood of each of its points.

Note that the intersection of any two neighborhoods of x in X is a neighborhood of x , which follows from the axiom that the intersection of two open sets is open.

The intuitive notion of “smallness” of a neighborhood is given by the concept of a neighborhood basis at a point:

2.4. Definition. If X is a topological space and $x \in X$ then a collection $\mathbf{B}_x$ of subsets of X containing x is called a *neighborhood basis* at x in X if each neighborhood of x in X contains some element of $\mathbf{B}_x$ and each element of $\mathbf{B}_x$ is a neighborhood of x .

Neighborhood bases are sometimes convenient in proving functions to be continuous:

2.5. Definition. A function $f: X \rightarrow Y$ between topological spaces is said to be *continuous at* x , where $x \in X$, if, given any neighborhood N of $f(x)$ in Y , there is a neighborhood M of x in X such that $f(M) \subset N$.

Since $f(f^{-1}(N)) \subset N$, this is the same as saying that $f^{-1}(N)$ is a neighborhood of x , for each neighborhood N of $f(x)$. Clearly, this need only be checked for N belonging to some neighborhood basis at $f(x)$.

2.6. Proposition. A function $f: X \rightarrow Y$ between topological spaces is continuous $\Leftrightarrow$ it is continuous at each point $x \in X$.

PROOF. Suppose that f is continuous, i.e., that $f^{-1}(U)$ is open for each open $U \subset Y$. Let N be a neighborhood of $f(x)$ in Y and let U be an open set such that $f(x) \in U \subset N$ as guaranteed by the definition of neighborhood. Then $x \in f^{-1}(U) \subset f^{-1}(N)$ and $f^{-1}(U)$ is open. It follows that $f^{-1}(N)$ is a neighborhood of x . Thus f is continuous at x .

Conversely, suppose that f is continuous at each point and let $U \subset Y$ be an open set. For any $x \in f^{-1}(U)$, $f^{-1}(U)$ is then a neighborhood of x . Thus there exists an open set V_x in X with $x \in V_x \subset f^{-1}(U)$. Hence $f^{-1}(U)$ is the union of the sets V_x for x ranging over $f^{-1}(U)$. Since the union of any collection of open sets is open, it follows that $f^{-1}(U)$ is open. But U was an arbitrary open set in Y and, consequently, f is continuous. $\square$

2.7. Definition. A function $f: X \rightarrow Y$ between topological spaces is called a *homeomorphism* if $f^{-1}: Y \rightarrow X$ exists (i.e., f is one-one and onto) and both f and f^{-1} are continuous. The notation $X \approx Y$ means that X is homeomorphic to Y .

Two topological spaces are, then, homeomorphic if there is a one-one correspondence between them as sets which also makes the open sets correspond. Homeomorphic spaces are considered as essentially the same. One of the main problems in topology is to find methods of deciding when two spaces are homeomorphic or not.

To describe a topological space it is not necessary to describe completely the open sets. This can often be done more simply using the notion of a “basis” for the topology:

2.8. Definition. If X is a topological space and $\mathbf{B}$ is a collection of subsets of X , then $\mathbf{B}$ is called a *basis* for the topology of X if the open sets are precisely the unions of members of $\mathbf{B}$. (In particular, the members of $\mathbf{B}$ are open.) A collection $\mathbf{S}$ of subsets of X is called a *subbasis* for the topology of X if the set $\mathbf{B}$ of *finite* intersections of members of $\mathbf{S}$ is a basis.

Note that *any* collection $\mathbf{S}$ of subsets of any set X is a subbasis for *some* topology on X , namely, the topology for which the open sets are the arbitrary unions of the finite intersections of members of $\mathbf{S}$. (The empty set and whole set X are taken care of by the convention that an intersection of an empty collection of sets is the whole set and the union of an empty collection of sets is the empty set.) Thus, to define a topology, it suffices to specify some collection of sets as a subbasis. The resulting topology is called the topology “generated” by this subbasis.

In a metric space the collection of ϵ -balls, for all $\epsilon > 0$, is a basis. So is the collection of ϵ -balls for $\epsilon = 1, \frac{1}{2}, \frac{1}{3}, \dots$.

Here are some examples of topological spaces:

1. (Trivial topology.) Any set X with only the empty set and the whole set X as open.
2. (Discrete topology.) Any set X with all subsets being open.
3. Any set X with open sets being those subsets of X whose complements are finite, together with the empty set. (That is, the closed sets are finite sets and X itself.)

4. $X = \omega \cup \{\omega\}$ with the open sets being all subsets of ω together with complements of finite sets. (Here, ω denotes the set of natural numbers.)
5. Let X be any partially ordered set. For $\alpha \in X$ consider the one-sided intervals $\{\beta \in X \mid \alpha < \beta\}$ and $\{\beta \in X \mid \alpha > \beta\}$. The “order topology” on X is the topology generated by these intervals. The “strong order topology” is the topology generated by these intervals together with the complements of finite sets.
6. Let $X = I \times I$ where I is the unit interval $[0, 1]$. Give this the “dictionary ordering,” i.e., $(x, y) < (s, t) \Leftrightarrow$ either $x < s$ or $(x = s \text{ and } y < t)$. Let X have the order topology for this ordering.
7. Let X be the real line but with the topology generated by the “half open intervals” $[x, y)$. This is called the “half open interval topology.”
8. Let $X = \Omega \cup \{\Omega\}$ be the set of ordinal numbers up to and including the least uncountable ordinal Ω ; see Theorem B.28. Give it the order topology.

2.9. Definition. A topological space is said to be *first countable* if each point has a countable neighborhood basis.

2.10. Definition. A topological space is said to be *second countable* if its topology has a countable basis.

Note that all metric spaces are first countable. Some metric spaces are not second countable, e.g., the space consisting of any uncountable set with the metric $\text{dist}(x, y) = 1$ if $x \neq y$, and $\text{dist}(x, x) = 0$ (which yields the discrete topology).

Euclidean spaces are second countable since the ϵ -balls, with ϵ rational, about the points with all rational coordinates, is easily seen to be a basis.

2.11. Definition. A sequence $f_1, f_2, \dots$ of functions from a topological space X to a metric space Y is said to *converge uniformly* to a function $f: X \rightarrow Y$ if, for each $\epsilon > 0$, there is a number n such that $i > n \Rightarrow \text{dist}(f_i(x), f(x)) < \epsilon$ for all $x \in X$.

2.12. Theorem. If a sequence $f_1, f_2, \dots$ of continuous functions from a topological space X to a metric space Y converges uniformly to a function $f: X \rightarrow Y$, then f is continuous.

PROOF. Given $\epsilon > 0$, let n_0 be such that

$$n \geq n_0 \Rightarrow \text{dist}(f(x), f_n(x)) < \epsilon/3 \quad \text{for all } x \in X.$$

Given a point x_0 , the continuity of f_{n_0} implies that there is a neighborhood N of x_0 such that $x \in N \Rightarrow \text{dist}(f_{n_0}(x), f_{n_0}(x_0)) < \epsilon/3$. Thus, for any $x \in N$ we have

$$\begin{aligned} \text{dist}(f(x), f(x_0)) &\leq \text{dist}(f(x), f_{n_0}(x)) + \text{dist}(f_{n_0}(x), f_{n_0}(x_0)) + \text{dist}(f_{n_0}(x_0), f(x_0)) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon. \end{aligned} \quad \square$$

2.13. Definition. A function $f: X \rightarrow Y$ between topological spaces is said to be *open* if $f(U)$ is open in Y for all open $U \subset X$. It is said to be *closed* if $f(C)$ is closed in Y for all closed $C \subset X$.

2.14. Definition. If X is a set and some condition is given on subsets of X which may or may not hold for any particular subset, then if there is a topology T whose open sets satisfy the condition, and such that, for any topology T' whose open sets satisfy the condition, then the T -open sets are also T' -open (i.e., $T \subset T'$), then T is called the *smallest* (or *weakest* or *coarsest*) topology satisfying the condition. If, instead, for any topology T' whose open sets satisfy the condition, any T' -open sets are also T -open, then T is called the *largest* (or *strongest* or *finest*) topology satisfying the condition.

The terms “weak” and “strong” are the oldest historically. However, they are used in some places to mean the opposite of the above meaning in general topology. Even some topology books disagree on their meaning. For this reason, the terms “coarse” and “fine” were introduced to rectify the confusion. They are metaphors for thinking of open sets as grains in a rock (the fewer grains, the coarser the rock). The terms “smallest” and “largest” were introduced for the same reason, and they are mathematically more precise as applied to the topologies as collections of open sets. We prefer the latter terms in general.

For example (see Section 13), if $f: X \rightarrow Y$ is a function and X is a topological space, then there is a largest topology on Y making f continuous, namely that topology having open sets $\{V \subset Y \mid f^{-1}(V) \text{ is open in } X\}$. There is also a smallest such topology, the trivial topology, but it is not very interesting. Also see Sections 8 and 13 for other examples of this concept.

If a topology is the largest one satisfying some given condition then usually (in fact, always) there is another condition for which the given topology is the smallest one satisfying the new condition. For example, the topology on Y , in the example of the previous paragraph, is the smallest topology satisfying the condition “for all spaces Z and all functions $g: Y \rightarrow Z$, $g \circ f$ continuous $\Rightarrow g$ continuous.” Thus it is meaningless to argue whether a given topology is “weak” or “strong,” etc., unless the defining condition is specified.

PROBLEMS

1. $\circledast$ Show that in a topological space X :
 - (a) the union of two closed sets is closed;
 - (b) the intersection of any collection of closed sets is closed; and
 - (c) the empty set $\emptyset$ and whole space X are closed.
2. Consider the topology on the real line generated by the half open intervals $[x, y)$ together with those of the form $(x, y]$. Show that this coincides with the discrete topology.
3. Show that the space $\Omega \cup \{\Omega\}$ in the order topology cannot be given a metric consistent with its topology.

4. $\diamond$ If $f: X \rightarrow Y$ is a function between topological spaces, and $f^{-1}(U)$ is open for each open U in some subbasis for the topology of Y , show that f is continuous.
5. $\diamond$ Suppose that S is a set and that we are given, for each $x \in S$, a collection $\mathbf{N}(x)$ of subsets of S satisfying:
- (1) $N \in \mathbf{N}(x) \Rightarrow x \in N$;
 - (2) $N, M \in \mathbf{N}(x) \Rightarrow \exists P \in \mathbf{N}(x) \ni P \subset N \cap M$; and
 - (3) $x \in S \Rightarrow \mathbf{N}(x) \neq \emptyset$.

Then show that there is a unique topology on S such that $\mathbf{N}(x)$ is a neighborhood basis at x , for each $x \in S$. (Thus a topology can be defined by the specification of such a collection of neighborhoods at each point.)

3. Subspaces

There are several techniques for producing new topological spaces out of old ones. The simplest is the passing to a “subspace,” which is merely an arbitrary subset inheriting a topology from the mother space in a quite natural way.

3.1. Definition. If X is a topological space and $A \subset X$ then the *relative topology* or the *subspace topology* on A is the collection of intersections of A with open sets of X . With this topology, A is called a *subspace* of X .

The following propositions are all easy consequences of the definitions and the proofs are left to the reader:

3.2. Proposition. If Y is a subspace of X then $A \subset Y$ is closed in $Y \Leftrightarrow A = Y \cap B$ for some closed subset B of X . □

3.3. Proposition. If X is a topological space and $A \subset X$ then there is a largest open set U with $U \subset A$. This set is called the “interior” of A in X and is denoted by $\text{int}(A)$. □

3.4. Proposition. If X is a topological space and $A \subset X$ then there is a smallest closed set F with $A \subset F \subset X$. This set is called the “closure” of A in X and is denoted by $\bar{A}$. □

If we need to specify the space in which a closure is taken (the X), we shall use the notation $\bar{A}^X$. A consequence of the following fact is that this notation need not be used very often:

3.5. Proposition. If $A \subset Y \subset X$ then $\bar{A}^Y = \bar{A}^X \cap Y$. Thus, if Y is closed in X then $\bar{A}^Y = \bar{A}^X$. □

3.6. Definition. If X is a topological space and $A \subset X$ then the *boundary* or *frontier* of A is defined to be $\partial A = \text{bdry}(A) = \bar{A} \cap X - A$.

3.7. Proposition. If $Y \subset X$ then the set of intersections of Y with members of a basis of X is a basis of the relative topology of Y . $\square$

3.8. Proposition. If X, Y, Z are topological spaces and Y is a subspace of X and Z is a subspace of Y , then Z is a subspace of X . $\square$

3.9. Proposition. If X is a metric space and $A \subset X$ then $\bar{A}$ coincides with the set of limits in X of sequences of points in A .

PROOF. If x is the limit of a sequence of points in A then any open set about x contains a point of A . Thus $x \notin \text{int}(X - A)$. Since $X - \text{int}(X - A) = \bar{A}$ (see the problems at the end of this section), $x \in \bar{A}$. Conversely, if $x \in \bar{A}$ and $n > 0$ is any integer, then $B_{1/n}(x)$ must contain a point in A because otherwise x would lie in $\text{int}(X - A)$. Take one such point and name it x_n . Then it follows immediately that $x = \lim(x_n)$ is a limit of a sequence of points in A . $\square$

3.10. Definition. A subset A of a topological space X is called *dense* in X if $\bar{A} = X$. A subset A is said to be *nowhere dense* in X if $\text{int}(\bar{A}) = \emptyset$.

PROBLEMS

1. $\star$ Let X be a topological space and $A, B \subset X$.

(a) Show that

$$\text{int}(A) = \{a \in X \mid \exists U \text{ open } \ni a \in U \subset A\}$$

and

$$\bar{A} = \{x \in X \mid \forall U \text{ open with } x \in U, U \cap A \neq \emptyset\}.$$

(b) Show that A is open $\Leftrightarrow A = \text{int}(A)$ and that A is closed $\Leftrightarrow A = \bar{A}$.

(c) Show that $X - \text{int}(A) = \overline{X - A}$ and that $X - \bar{A} = \text{int}(X - A)$.

(d) Show that $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$ and that $\bar{A} \cup \bar{B} = \overline{A \cup B}$.

(e) Show that

$$\bigcap \text{int}(A_\alpha) \supset \text{int}(\bigcap A_\alpha) = \text{int}(\bigcap \text{int}(A_\alpha)),$$

$$\bigcup \bar{A}_\alpha \subset \text{closure}(\bigcup A_\alpha) = \text{closure}(\bigcup \bar{A}_\alpha),$$

$$\bigcup \text{int}(A_\alpha) \subset \text{int}(\bigcup A_\alpha),$$

$$\bigcap \bar{A}_\alpha \supset \text{closure}(\bigcap A_\alpha),$$

and give examples showing that these inclusions need not be equalities.

(f) Show $A \subset B \Rightarrow [\bar{A} \subset \bar{B} \text{ and } \text{int}(A) \subset \text{int}(B)]$.

2. $\star$ For $A \subset X$, a topological space, show that X is the *disjoint* union of $\text{int}(A)$, $\text{bdry}(A)$, and $X - \bar{A}$.

3. $\star$ Show that a metric space is second countable $\Leftrightarrow$ it has a countable dense set (a countable set whose closure is the whole space). (Such a metric space is called “separable.”)

4. $\diamond$ Show that the union of two nowhere dense sets is nowhere dense.
5. A topological space X is said to be “irreducible” if, whenever $X = F \cup G$ with F and G closed, then either $X = F$ or $X = G$. A subspace is irreducible if it is so in the subspace topology. Show that if X is irreducible and $U \subset X$ is open, then U is irreducible.
6. A “Zariski space” is a topological space with the property that every descending chain $F_1 \supset F_2 \supset F_3 \supset \cdots$ of closed sets is eventually constant. Show that every Zariski space can be expressed as a finite union $X = Y_1 \cup Y_2 \cup \cdots \cup Y_n$ where the Y_i are closed and irreducible and $Y_i \not\subset Y_j$ for $i \neq j$. Also show that this decomposition is unique up to order.
7. Let X be the real line with the topology for which the open sets are $\emptyset$ together with the complements of finite subsets. Show that X is an irreducible Zariski space.
8. $\diamond$ Let $X = A \cup B$, where A and B are closed. Let $f: X \rightarrow Y$ be a function. If the restrictions of f to A and B are both continuous then show that f is continuous.

4. Connectivity and Components

In a naively intuitive sense, a connected space is a space in which one can move from any point to any other point without jumps. Another way to view it intuitively is as the idea that the space does not fall into two or more pieces which are separated from one another. There are two ways of making these crude ideas precise and both of them will be important to us. One of them, called “connectivity,” is the subject of this section, while the other, called “arcwise connectivity,” is taken up in the problems at the end.

4.1. Definition. A topological space X is called *connected* if it is not the disjoint union of two nonempty open subsets.

4.2. Definition. A subset A of a topological space X is called *clopen* if it is both open and closed in X .

4.3. Proposition. A topological space X is connected $\Leftrightarrow$ its only clopen subsets are X and $\emptyset$. □

4.4. Definition. A *discrete valued map* is a map (continuous) from a topological space X to a discrete space D .

4.5. Proposition. A topological space X is connected $\Leftrightarrow$ every discrete valued map on X is constant.

PROOF. If X is connected and $d: X \rightarrow D$ is a discrete valued map and if $y \in D$ is in the range of d , then $d^{-1}(y)$ is clopen and nonempty and so must equal X , and so d is constant with only value y .

Conversely, if X is not connected then $X = U \cup V$ for some disjoint clopen sets U and V . Then the map $d: X \rightarrow \{0, 1\}$ which is 0 on U and is 1 on V is a nonconstant discrete valued map. $\square$

4.6. Proposition. *If $f: X \rightarrow Y$ is continuous and X is connected, then $f(X)$ is connected.*

PROOF. Let $d: f(X) \rightarrow D$ be a discrete valued map. Then $d \circ f$ is a discrete valued map on X and hence must be constant. But that implies that d is constant, and hence that $f(X)$ is connected. $\square$

4.7. Proposition. *If $\{Y_i\}$ is a collection of connected sets in a topological space X and if no two of the Y_i are disjoint, then $\bigcup Y_i$ is connected.*

PROOF. Let $d: \bigcup Y_i \rightarrow D$ be a discrete valued map. Let p, q be any two points in $\bigcup Y_i$. Suppose $p \in Y_i$ and $q \in Y_j$ and $r \in Y_i \cap Y_j$. Then, since d must be constant on each Y_i , we have $d(p) = d(r) = d(q)$. But p and q were completely arbitrary. Thus d is constant. $\square$

4.8. Corollary. *The relation “ p and q belong to a connected subset of X ” is an equivalence relation.* $\square$

4.9. Definition. The equivalence classes of the equivalence relation in Corollary 4.8 are called the *components* of X .

4.10. Proposition. *Components of space X are connected and closed. Each connected set is contained in a component. (Thus the components are “maximal connected subsets.”) Components are either equal or disjoint, and fill out X .*

PROOF. The last statement follows from the fact that the components are equivalence classes of an equivalence relation. By definition, the component of X containing p is the union of all connected sets containing p , and that is connected by Proposition 4.7. This also implies that a connected set lies in a component. That a component is closed follows from the fact that the closure of a connected set is connected (left to the reader in the problems below). $\square$

4.11. Proposition. *The statement “ $d(p) = d(q)$ for every discrete valued map d on X ” is an equivalence relation.* $\square$

4.12. Definition. The equivalence classes of the relation in Proposition 4.11 are called the *quasi-components* of X .

4.13. Proposition. *Quasi-components of a space X are closed. Each connected set is contained in a quasi-component. (In particular, each component is contained in a quasi-component.) Quasi-components are either equal or disjoint, and fill out X .*

PROOF. If $p \in X$ then the quasi-component containing it is just

$$\{q \in X \mid d(q) = d(p) \text{ for all discrete valued maps } d\}.$$

But this is

$$\bigcap \{d^{-1}(d(p)) \mid d \text{ a discrete valued map}\}$$

which is an intersection of closed sets and hence is closed. The rest is obvious. $\square$

PROBLEMS

1. $\star$ If A is a connected subset of the topological space X and if $A \subset B \subset \bar{A}$ then show that B is connected.
2. $\star$ A space X is said to be “locally connected” if for each $x \in X$ and each neighborhood N of x , there is a connected neighborhood V of x with $V \subset N$. If X is locally connected, show that its components are open and equal its quasi-components.
3. $\star$ Show that the unit interval $[0, 1]$ in the real number is connected. (*Hint:* Assume that $[0, 1] = U \cup V$, where U and V are disjoint nonempty open sets, and $1 \in V$. Consider $x = \sup(U)$. Show that $x < 1$ and derive a contradiction.)
4. Consider the subspace X of the unit square in the plane consisting of the vertical line segments $\{1/n\} \times [0, 1]$ for $n = 1, 2, 3, \dots$, and the two points $(0, 0)$ and $(0, 1)$. Show that the latter two points are components of X but not quasi-components. Show that the two point set $\{(0, 0), (0, 1)\}$ is a quasi-component which is not connected.
5. $\star$ A topological space X is said to be “arcwise connected” if for any two points p and q in X there exists a map $\lambda: [0, 1] \rightarrow X$ with $\lambda(0) = p$ and $\lambda(1) = q$. A space X is “locally arcwise connected” if every neighborhood of any point contains an arcwise connected neighborhood. An “arc component” is a maximal arcwise connected subset. Show that:
 - (a) an arcwise connected space is connected;
 - (b) a space is the disjoint union of its arc components;
 - (c) an arc component of a space is contained in some component;
 - (d) the arc components of a locally arcwise connected space are clopen, and coincide with the components;
 - (e) the space with exactly two points p and q and open sets $\emptyset, \{p\}, \{p, q\}$ (only) is arcwise connected; and
 - (f) the subspace of the plane consisting of $\{0\} \times [-1, 1] \cup \{(x, \sin(1/x)) \mid x > 0\}$ is connected but not arcwise connected.

5. Separation Axioms

The axioms defining a topological space are extremely general and weak. It should be no surprise that most spaces of interest will have further restrictions on them. We refer here not to structures like a metric, but to conditions

completely describable in terms of the topology itself, i.e., in terms of the points and open sets. We begin with the so-called separation axioms.

5.1. Definition. The separation axioms:

- (T₀) A topological space X is called a T_0 -space if for any two points $x \neq y$ there is an open set containing one of them but not the other.
- (T₁) A topological space X is called a T_1 -space if for any two points $x \neq y$ there is an open set containing x but not y and another open set containing y but not x .
- (T₂) A topological space X is called a T_2 -space or *Hausdorff* if for any two points $x \neq y$ there are disjoint open sets U and V with $x \in U$ and $y \in V$.
- (T₃) A T_1 -space X is called a T_3 -space or *regular* if for any point x and closed set F not containing x there are disjoint open sets U and V with $x \in U$ and $F \subset V$.
- (T₄) A T_1 -space X is called a T_4 -space or *normal* if for any two disjoint closed sets F and G there are disjoint open sets U and V with $F \subset U$ and $G \subset V$.

Axiom T₀ simply says that points can be distinguished by the open sets in which they lie.

Axiom T₁ is the same as saying that one-point sets (singletons) are closed sets, because if we single out a point x and, for each different point y we take U_y to be an open set containing y but not x , then $X - \{x\} = \bigcup U_y$ is the union of open sets and so is open. Conversely, if $\{x\}$ is closed then the open set $X - \{x\}$ can be taken, in the axiom, as the open set containing any other point.

Axiom T₂ is the most important of these axioms and will be assumed in the majority of the text of this book. We shall see later that it essentially means that “limits” are unique.

5.2. Proposition. *A Hausdorff space is regular $\Leftrightarrow$ the closed neighborhoods of any point form a neighborhood basis of the point.*

PROOF. Suppose that X is regular, let $x \in V$, with V open, and put $C = X - V$. By regularity there are open sets U , W , with $x \in U$, $C \subset W$, and $U \cap W = \emptyset$. Then $X - W$ is closed, and we have $X - W \subset X - C = V$, so any neighborhood V of x contains a closed neighborhood $X - W$ of x , as was to be shown.

Conversely, suppose that every point has a closed neighborhood basis. Let $x \notin C$ with C closed and put $V = X - C$. By the assumption, there is an open set U with $\bar{U} \subset V = X - C$ and $x \in U$. Then $C \subset X - \bar{U}$, and $U \cap (X - \bar{U}) = \emptyset$. Thus X is regular. $\square$

5.3. Corollary. *A subspace of a regular space is regular.*

PROOF. If $A \subset X$ is a subspace, just intersect a closed neighborhood basis in

X of a point $a \in A$ with A and you get a closed neighborhood basis of a in A . $\square$

PROBLEMS

1. Give an example of a space that is not T_0 , and an example of a T_0 -space that is not T_1 . (*Hint*: Spaces with only two points suffice.)
2. Show that a finite T_1 -space is discrete.
3. Consider the set ω of natural numbers together with two other points named x, y . Put a partial ordering on this set which orders ω as usual and makes both x and y greater than any integer, but does not order x against y . Give this the strong order topology. Show it is T_1 but not Hausdorff.
4. Consider the space X whose point set is the plane but whose open sets are given by the basis consisting of the usual open sets in the plane together with the sets $\{(x, y) | x^2 + y^2 < a, y \neq 0\} \cup \{(0, 0)\}$ for all $a > 0$. Show that X is Hausdorff but not regular.
5. $\diamond$ Show that a subspace of a Hausdorff space is Hausdorff.
6. $\diamond$ Show that a Hausdorff space is normal $\Leftrightarrow$ for any sets U open and C closed with $C \subset U$ there is an open set V with $C \subset V \subset \bar{V} \subset U$.
7. Show that there is a smallest topology on the real numbers such that every singleton is closed. Which of the separation axioms does it satisfy?
8. Show that if a Zariski space (see Section 3, Problem 6) is Hausdorff then it is finite.
9. $\blacklozenge \diamond$ Show that a metric space is normal.

6. Nets (Moore–Smith Convergence) $\odot$

In metric spaces continuity of functions can be expressed in terms of the convergence of sequences. This is not true in general topological spaces. However, there is a generalization of sequences that does work and permits proofs of some things analogously to proofs using sequences in metric spaces. This can be of great help to the intuition. The generalization of a sequence is called a net, and we will develop this subject in this section. Although we will use this concept in proving a couple of important results in subsequent sections, those results will not be used in the main body of the book, and for that reason, this section can be skipped without serious harm to subsequent developments.

6.1. Definition. A *directed set* D is a partially ordered set such that, for any two elements α and β of D , there is a $\tau \in D$ with $\tau \geq \alpha$ and $\tau \geq \beta$.

6.2. Definition. A *net* in a topological space X is a directed set D together with a function $\Phi: D \rightarrow X$.

Note that a sequence is simply a net based on the natural numbers as indexing set.

6.3. Definition. If $\Phi: D \rightarrow X$ is a net in the topological space X and $A \subset X$ then we say that Φ is *frequently* in A if for any $\alpha \in D$ there is a $\beta \geq \alpha$ such that $\Phi(\beta) \in A$. It is said to be *eventually* in A if there is an $\alpha \in D$ such that $\Phi(\beta) \in A$ for all $\beta \geq \alpha$.

6.4. Definition. A net $\Phi: D \rightarrow X$ in a topological space is said to converge to $x \in X$ if, for every neighborhood $U \subset X$ of x , Φ is eventually in U .

Note that if a net Φ is eventually in two sets U and V then it is eventually in $U \cap V$. Also, this is impossible if $U \cap V = \emptyset$. This proves half of the following fact. The remainder of the proof constructs a net which is typical of the nets encountered with general topological spaces.

6.5. Proposition. *A topological space X is Hausdorff $\Leftrightarrow$ any two limits of any convergent net are equal. (Thus one can speak of the limit of a net in such a space.)*

PROOF. The implication $\Rightarrow$ follows from the preceding discussion. Thus suppose that X is not Hausdorff, and that $x, y \in X$ are two points which cannot be separated by open sets. Consider the directed set whose elements are ordered pairs $\alpha = \langle U, V \rangle$ of open sets where $x \in U$ and $y \in V$ with the ordering $\langle U, V \rangle \geq \langle A, B \rangle \Leftrightarrow (U \subset A \text{ and } V \subset B)$. For any $\alpha = \langle U, V \rangle$, let $\Phi(\alpha)$ be some point in $U \cap V$. This defines a net Φ which we claim converges to both x and y .

To see this, let W be any neighborhood of x . We claim that Φ is eventually in W . In fact, take any open set V containing y and an open set U with $x \in U \subset W$ and let $\alpha = \langle U, V \rangle$. If $\beta = \langle A, B \rangle \geq \alpha$ then $A \subset U$ and $B \subset V$ so that $\Phi(\beta) \in A \cap B \subset U \subset W$, as claimed. Thus Φ converges to x . Similarly, it converges to y . $\square$

Next we show that nets are “sufficient” to describe continuity.

6.6. Proposition. *A function $f: X \rightarrow Y$ between two topological spaces is continuous $\Leftrightarrow$ for every net Φ in X converging to $x \in X$, the net $f \circ \Phi$ in Y converges to $f(x)$.*

PROOF. First suppose that f is continuous and let Φ be a net in X converging to x . Let V be any open set in Y containing $f(x)$ and put $U = f^{-1}(V)$, which is a neighborhood of x . By definition of convergence, Φ is eventually in U , and so $f \circ \Phi$ is eventually in V , and thus converges to $f(x)$.

Conversely, suppose that f is not continuous. Then there is an open set $V \subset Y$ such that $K = f^{-1}(V)$ is not open. Let $x \in K - \text{int}(K)$. Consider the directed set consisting of open neighborhoods of x ordered by inclusion, i.e.,

$A \leq B$ means $A \supset B$. For any such neighborhood A of x , A cannot be completely inside K , so we can choose a point $w_A \in A - K$. Define the net Φ by putting $\Phi(A) = w_A$. If N is any neighborhood of x and if $B \geq N$ (i.e., $B \subset N$) then $\Phi(B) = w_B \in B - K \subset N$, showing that Φ is eventually in N . Thus Φ converges to x . However $(f \circ \Phi)(A) \notin V$, for any A , so that $f \circ \Phi$ is not eventually in V , and thus does not converge to $f(x)$. $\square$

Given a particular net $\Phi: D \rightarrow X$ let $x_\alpha = \Phi(\alpha)$, for $\alpha \in D$. Then it is common to speak of $\{x_\alpha\}$ as being the net in question. This notation makes discussion of nets similar to the notation commonly used with sequences. For example, one can phrase the condition in Proposition 6.6 as

$$f(\lim x_\alpha) = \lim(f(x_\alpha)).$$

6.7. Proposition. *If $A \subset X$ then $\bar{A}$ coincides with the set of limits of nets in A which converge in X .*

PROOF. If $x \in \bar{A}$ then any open neighborhood U of x must intersect A nontrivially. Thus we can base a net on this set of neighborhoods, ordered by inclusion and such points $x_U \in U \cap A$. This clearly converges to x . Conversely, if $\{x_\alpha\}$ is any net of points in A which converges to a point $x \in X$ then, by definition, this net is eventually in any given neighborhood of x . Thus any neighborhood of x contains a point in A and so $x \in \bar{A}$. (Here we are using Problem 1(a) of Section 3.) $\square$

In the case of ordinary sequences, a subsequence can be thought of in two different ways: (1) by discarding elements of the sequence and renumbering, or (2) by composing the sequence, thought of as a function $\mathbf{Z}^+ \rightarrow X$, with a function $h: \mathbf{Z}^+ \rightarrow \mathbf{Z}^+$, such that $i > j \Rightarrow h(i) > h(j)$. The first of these turns out to be inadequate for nets in general spaces. For the second method, a little thought should convince the reader that the last condition of monotonicity of h is stronger than is necessary for the usual uses of subsequences. Modifying it leads to the more general notion of a "subnet," which we now define.

6.8. Definition. If D and D' are directed sets and $h: D' \rightarrow D$ is a function, then h is called *final* if, $\forall \delta \in D, \exists \delta' \in D' \ni (\alpha' \geq \delta' \Rightarrow h(\alpha') \geq \delta)$.

6.9. Definition. A *subnet* of a net $\mu: D \rightarrow X$, is the composition $\mu \circ h$ of μ with a final function $h: D' \rightarrow D$.

6.10. Proposition. *A net $\{x_\alpha\}$ is frequently in each neighborhood of a given point $x \in X \Leftrightarrow$ it has a subnet which converges to x .*

PROOF. Consider the directed set D' consisting of ordered pairs (α, U) where $\alpha \in D, U$ is a neighborhood of x , and $x_\alpha \in U$, ordered by the D ordering and

inclusion. If (α, U) and (β, V) are in D' then, since $\{x_\alpha\}$ is frequently in $U \cap V$, there is a $\gamma \geq \alpha, \beta$ with $x_\gamma \in U \cap V$. Thus $(\gamma, U \cap V) \in D'$ and $(\gamma, U \cap V) \geq (\alpha, U), (\beta, V)$, showing that D' is directed. Map $D' \rightarrow D$ by $(\alpha, U) \mapsto \alpha$. For any $\delta \in D$, we have $(\delta, X) \in D'$. Now $(\alpha, U) \geq (\delta, X)$ implies that $\alpha \geq \delta$, which means that $D' \rightarrow D$ is final, and so $\{x_{(\alpha, U)}\}$ is a subnet of $\{x_\alpha\}$. We claim that it converges to x . Let N be any neighborhood of x . By assumption, there is some $x_\beta \in N$. If $(\alpha, U) \geq (\beta, N)$ then $x_{(\alpha, U)} = x_\alpha \in U \subset N$. Consequently, $\{x_{(\alpha, U)}\}$ is eventually in N . The converse is immediate. $\square$

Next we treat a powerful concept for nets which has no analogue for sequences.

6.11. Definition. A net in a set X is called *universal* if, for any $A \subset X$, the net is either eventually in A or eventually in $X - A$.

6.12. Proposition. *The composition of a universal net in X with a function $f: X \rightarrow Y$ is a universal net in Y .*

PROOF. If $A \subset Y$ then the net is eventually in either $f^{-1}(A)$ or $X - f^{-1}(A)$ by definition. But $X - f^{-1}(A) = f^{-1}(Y - A)$ and it follows that the composed net is eventually in either A or $Y - A$, respectively. $\square$

Except for somewhat trivial cases, the definition of a universal net may seem so strong that the reader may reasonably doubt the existence of universal nets. However:

6.13. Theorem. *Every net has a universal subnet.*

PROOF. Let $\{x_\alpha | \alpha \in P\}$ be a net in X . Consider all collections $\mathbf{C}$ of subsets of X such that:

- (1) $A \in \mathbf{C} \Rightarrow \{x_\alpha\}$ is frequently in A ; and
- (2) $A, B \in \mathbf{C} \Rightarrow A \cap B \in \mathbf{C}$.

For example, $\mathbf{C} = \{X\}$ is such a collection. Order the family of all such collections $\mathbf{C}$ by inclusion. The union of any simply ordered set of such collections is clearly such a collection, i.e., satisfies (1) and (2). By the Maximality Principle, there is a maximal such collection $\mathbf{C}_0$.

Let $P_0 = \{(A, \alpha) \in \mathbf{C}_0 \times P | x_\alpha \in A\}$ and order P_0 by

$$(B, \beta) \geq (A, \alpha) \Leftrightarrow B \subset A \text{ and } \beta \geq \alpha.$$

This gives a partial order on P_0 making P_0 into a directed set. Map $P_0 \rightarrow P$ by taking (A, α) to α . This is clearly final and thus defines a subnet we shall denote by $\{x_{(A, \alpha)}\}$. We claim that this subnet is universal.

Suppose S is any subset of X such that $\{x_{(A, \alpha)}\}$ is frequently in S . Then, for any $(A, \alpha) \in P_0$, there is a $(B, \beta) \geq (A, \alpha)$ in P_0 with $x_\beta = x_{(B, \beta)} \in S$. Then $B \subset A$, $\beta \geq \alpha$, and $x_\beta \in B$. Thus $x_\beta \in S \cap B \subset S \cap A$. We conclude that $\{x_\alpha\}$ is

frequently in $S \cap A$, for any $A \in \mathbf{C}_0$. But then we can throw S and all the sets $S \cap A$, for $A \in \mathbf{C}_0$, into $\mathbf{C}_0$ and conditions (1) and (2) will still hold. By maximality, we must have $S \in \mathbf{C}_0$. If $\{x_{(A,a)}\}$ were *also* frequently in $X - S$ then $X - S$ would be in $\mathbf{C}_0$, and so $\emptyset = S \cap (X - S)$ would be in $\mathbf{C}_0$, by (2), and this is contrary to (1). Thus we conclude that $\{x_{(A,a)}\}$ is not frequently in $X - S$, and so is eventually in S .

We have shown that if $\{x_{(A,a)}\}$ is frequently in a set S then, in fact, it is eventually in S . This implies that $\{x_{(A,a)}\}$ is universal. $\square$

Note that this proof uses the Axiom of Choice in the guise of the Maximality Principle. In fact, it can be shown that Theorem 6.13 is equivalent to the Axiom of Choice.

The following fact is immediate from the definitions:

6.14. Proposition. *A subnet of a universal net is universal.* $\square$

PROBLEMS

1. Show that a sequence is a universal net if and only if it is eventually constant.
2. Consider the space $X = \Omega \cup \{\Omega\}$ of ordinals up to and including the first uncountable ordinal Ω with the order topology. Show explicitly that there is a net in Ω which converges to $\{\Omega\}$ but that there is no *sequence* which does so.
3. Prove Proposition 6.14.
4. $\blacklozenge$ Let H be a dense set in the topological space X and let $f: H \rightarrow Y$ be a map with Y regular. Let $g: X \rightarrow Y$ be a *function*. Suppose that for any net $\{h_x\}$ in H with $h_x \rightarrow x \in X$ we have $f(h_x) \rightarrow g(x)$. Then show that $g: X \rightarrow Y$ is continuous. Also show that the condition of regularity on Y is needed by giving a counterexample without it.

7. Compactness

The notion of compactness is one of the most important ideas in mathematics. The reader has undoubtedly already met it in connection with some of the fundamental facts about the real numbers used in calculus.

7.1. Definition. A *covering* of a topological space X is a collection of sets whose union is X . It is an *open covering* if the sets are open. A *subcover* is a subset of this collection which still covers the space.

If $A \subset X$ then, for convenience, we sometimes use “cover A ” for a collection of subsets of X whose union contains A .

7.2. Definition. A topological space X is said to be *compact* if every open

covering of X has a finite subcover. (This is sometimes referred to as the *Heine–Borel property*.)

7.3. Definition. A collection C of sets has the *finite intersection property* if the intersection of any finite subcollection is nonempty.

The following fact is just a simple translation of the definition of compactness in terms of open sets to a statement about the (closed) complements of those sets:

7.4. Theorem. *A topological space X is compact $\Leftrightarrow$ for every collection of closed subsets of X which has the finite intersection property, the intersection of the entire collection is nonempty.* $\square$

7.5. Theorem. *If X is a Hausdorff space, then any compact subset of X is closed.*

PROOF. Let $A \subset X$ be compact and suppose $x \in X - A$. For $a \in A$ let U_a and V_a be open sets with $U_a \cap V_a = \emptyset$. Now $A = \bigcup (U_a \cap A)$, which implies, by compactness of A , that there are $a_1, a_2, \dots, a_n \in A$, such that $A \subset U_{a_1} \cup \dots \cup U_{a_n} = U$. But $x \in V_{a_1} \cap \dots \cap V_{a_n} = V$, which is open, and $U \cap V = \emptyset$. Thus $x \in V \subset X - U \subset X - A$ and V is open. Since this is true for any $x \in X - A$, we conclude that $X - A$ is open, and so A is closed. $\square$

7.6. Theorem. *If X is compact and $f: X \rightarrow Y$ is continuous, then $f(X)$ is compact.*

PROOF. We may as well replace Y by $f(X)$ and so assume that f is onto. For any open cover of Y look at the inverse images of its sets and apply the compactness of X . $\square$

7.7. Theorem. *If X is compact, and $A \subset X$ is closed, then A is compact.*

PROOF. Cover A by open sets in X , throw in the open set $X - A$ and apply the compactness of X . $\square$

The following fact provides an easy way to check that certain constructions yield homeomorphisms, as we shall see:

7.8. Theorem. *If X is compact and Y is Hausdorff and $f: X \rightarrow Y$ is continuous, one–one, and onto, then f is a homeomorphism.*

PROOF. We are to show that f^{-1} is continuous. That is the same as showing that f is a closed mapping (takes closed sets to closed sets). But if $A \subset X$ is closed, then A is compact by Theorem 7.7, so $f(A)$ is compact by Theorem 7.6, whence $f(A)$ is closed by Theorem 7.5. $\square$

7.9. Theorem. *The unit interval $I = [0, 1]$ is compact.*

PROOF. Let $\mathbf{U}$ be an open covering of I . Put

$$S = \{s \in I \mid [0, s] \text{ is covered by a finite subcollection of } \mathbf{U}\}.$$

Let b be the least upper bound of S . Clearly S must be an interval of the form $S = [0, b)$ or $S = [0, b]$. In the former case, however, consider a set $U \in \mathbf{U}$ containing the point b . This set must contain an interval of the form $[a, b]$. But then we can throw U in with the hypothesized finite cover of $[0, a]$ to obtain a finite cover of $[0, b]$. Thus we must have that $S = [0, b]$ for some $b \in [0, 1]$. But if $b < 1$, then a similar argument shows that there is a finite cover of $[0, c]$ for some $c > b$, contradicting the choice of b . Thus $b = 1$ and we have found the desired finite cover of $[0, 1]$. $\square$

Note, of course, that any finite closed interval $[a, b]$ of real numbers is homeomorphic to $[0, 1]$ and hence is also compact. Any closed subset of $[a, b]$ is then compact. By looking at the covering of any subset of $\mathbf{R}$ by the intervals $(-n, n)$, we see that a compact set in $\mathbf{R}$ must be bounded. Consequently, a subset of $\mathbf{R}$ is compact $\Leftrightarrow$ it is closed and bounded. The reader is cautioned not to think that this holds in all metric spaces; see Corollary 8.7 and Theorem 9.4.

7.10. Theorem. *A real valued map on a compact space assumes a maximum value.*

PROOF. If $f: X \rightarrow \mathbf{R}$ is continuous and X is compact then $f(X)$ is compact by Theorem 7.6. Thus $f(X)$ is closed and bounded. Thus $\sup(f(X))$ exists, is finite, and belongs to $f(X)$ since $f(X)$ is closed. $\square$

7.11. Theorem. *A compact Hausdorff space is normal.*

PROOF. Suppose X is compact Hausdorff. We will first show that X is regular. For this, suppose C is a closed subset and $x \notin C$. Since X is Hausdorff, for any point $y \in C$ there are open sets U_y and V_y with $x \in U_y$, $y \in V_y$ and $U_y \cap V_y = \emptyset$. Since C is closed, it is compact, and the sets V_y cover it. Thus there are points $y_1, \dots, y_n$ so that $C \subset V_{y_1} \cup \dots \cup V_{y_n}$. If we put $U = U_{y_1} \cap \dots \cap U_{y_n}$ and $V = V_{y_1} \cup \dots \cup V_{y_n}$ then $x \in U$, $C \subset V$, and $U \cap V = \emptyset$ as desired. The remainder of the proof goes exactly the same way with C playing the role of x and the other closed set playing the role of C . $\square$

The following notion is mainly of use for locally compact spaces X, Y (see Section 11), but makes sense for all topological spaces:

7.12. Definition. A map $f: X \rightarrow Y$ between topological spaces is said to be *proper* if $f^{-1}(C)$ is compact for each compact subset C of Y .

7.13. Theorem. *If $f: X \rightarrow Y$ is a closed map and $f^{-1}(y)$ is compact for each $y \in Y$, then f is proper.*

PROOF. Let $C \subset Y$ be compact and let $\{U_\alpha | \alpha \in A\}$ be a collection of open sets whose union contains $f^{-1}(C)$. For any $y \in C$ there is a finite subset $A_y \subset A$ such that

$$f^{-1}(y) \subset \bigcup \{U_\alpha | \alpha \in A_y\}.$$

Put

$$W_y = \bigcup \{U_\alpha | \alpha \in A_y\}$$

and

$$V_y = Y - f(X - W_y),$$

which is open. Note that $f^{-1}(V_y) \subset W_y$ and $y \in V_y$. Since C is compact and is covered by the V_y , there are points $y_1, \dots, y_n$ such that $C \subset V_{y_1} \cup \dots \cup V_{y_n}$. Thus

$$\begin{aligned} f^{-1}(C) &\subset f^{-1}(V_{y_1}) \cup \dots \cup f^{-1}(V_{y_n}) \subset W_{y_1} \cup \dots \cup W_{y_n} \\ &= \bigcup \{U_\alpha | \alpha \in A_{y_i}; i = 1, 2, \dots, n\}, \end{aligned}$$

a finite union. □

7.14. Theorem. *For a topological space X the following are equivalent:*

- (1) X is compact.
- (2) Every collection of closed subsets of X with the finite intersection property has a nonempty intersection.
- (3) Every universal net in X converges.
- (4) Every net in X has a convergent subnet.

PROOF. We have already handled the equivalence of (1) and (2). For the rest:

(1) $\Rightarrow$ (3) Suppose $\{x_\alpha\}$ is a universal net that does not converge. Then given $x \in X$, there is an open neighborhood U_x of x such that x_α is not eventually in U_x . Then x_α is eventually in $X - U_x$ by definition of universal. That is, there is an index β_x such that $\alpha \geq \beta_x \Rightarrow x_\alpha \notin U_x$. Cover X by $U_{x_1} \cup \dots \cup U_{x_n}$. Let $\alpha \geq \beta_{x_i}$ for all i . Then $x_\alpha \notin U_{x_i}$ for any i , which means that $x_\alpha \notin X$, an absurdity.

(3) $\Rightarrow$ (4) is clear since every net has a universal subnet.

(4) $\Rightarrow$ (2) Let $\mathbf{F} = \{C\}$ be a collection of closed sets with the finite intersection property. We can throw in all finite intersections and so assume that $\mathbf{F}$ is closed under finite intersection. Then $\mathbf{F}$, ordered by $C \geq C' \Leftrightarrow C \subset C'$, is directed. For each $C \in \mathbf{F}$ let $x_C \in C$, defining a net. By assumption, there is a convergent subnet, given by a final map $f: D \rightarrow \mathbf{F}$, say. Thus, for $\alpha \in D$, $f(\alpha) \in \mathbf{F}$ and $x_{f(\alpha)} \in f(\alpha)$. Suppose $x_{f(\alpha)} \rightarrow x$. Let $C \in \mathbf{F}$. Then there is a $\beta \in D \ni \alpha \geq \beta \Rightarrow f(\alpha) \subset C$, and so $x_{f(\alpha)} \in f(\alpha) \subset C$. Since C is closed it follows from Proposition 6.7 that $x \in C$. Thus $x \in \bigcap \{C \in \mathbf{F}\}$, proving (2). □

PROBLEMS

1. Give a direct proof of (1) $\Rightarrow$ (4) in Theorem 7.14 without use of universal nets.
2. $\diamond$ Let X be a compact space and let $\{C_\alpha | \alpha \in A\}$ be a collection of closed sets, closed with respect to finite intersections. Let $C = \bigcap C_\alpha$ and suppose that $C \subset U$ with U open. Show that $C_\alpha \subset U$ for some α .

3. Give an example showing that the hypothesis, in Theorem 7.13, that f is closed, cannot be dropped.

8. Products

Let X and Y be topological spaces. Then we can define a topology (called the “product topology”) on $X \times Y$ by taking the collection of sets $U \times V$ to be a subbase, where $U \subset X$ and $V \subset Y$ are open. Since

$$U_1 \times V_1 \cap U_2 \times V_2 = (U_1 \cap U_2) \times (V_1 \cap V_2),$$

this is, in fact, a basis. Therefore the open sets are precisely the arbitrary unions of such “rectangles.”

Similarly we can define a product topology on finite products $X_1 \times X_2 \times \cdots \times X_n$ of topological spaces.

For an infinite product $\times \{X_\alpha | \alpha \in A\}$, we define the product topology as the topology with a basis consisting of the sets $\times \{U_\alpha | \alpha \in A\}$ where the U_α are open and where we demand that $U_\alpha = X_\alpha$ for all but a finite number of α 's. Note that the collection of sets of the form $U_\alpha \times \times \{X_\beta | \beta \neq \alpha\}$ is a subbasis for the product topology. This topology is also called the “Tychonoff topology.”

8.1. Proposition. *The projections $\pi_X: X \times Y \rightarrow X$ and $\pi_Y: X \times Y \rightarrow Y$ are continuous, and the product topology is the smallest topology for which this is true. Similarly for the case of infinite products.*

PROOF. The subbasis last described consists of exactly those sets which must be open for the projections to be continuous, and the proposition is just expressing that. $\square$

8.2. Proposition. *If X is compact then the projection $\pi_Y: X \times Y \rightarrow Y$ is closed.*

PROOF. Let $C \subset X \times Y$ be closed. We are to show that $Y - \pi_Y(C)$ is open. Let $y \notin \pi_Y(C)$, i.e., $\langle x, y \rangle \notin C$ for all $x \in X$. Then, for any $x \in X$, there are open sets $U_x \subset X$ and $V_x \subset Y$ such that $x \in U_x$, $y \in V_x$, and $(U_x \times V_x) \cap C = \emptyset$.

Since X is compact there are points $x_1, \dots, x_n \in X$ such that $U_{x_1} \cup \cdots \cup U_{x_n} = X$. Let $V = V_{x_1} \cap \cdots \cap V_{x_n}$. Then

$$(X \times V) \cap C = (U_{x_1} \cup \cdots \cup U_{x_n}) \times (V_{x_1} \cap \cdots \cap V_{x_n}) \cap C = \emptyset.$$

Thus, $y \in V \subset Y - \pi_Y(C)$ and V is open. Since y was an arbitrary point of $Y - \pi_Y(C)$ it follows that this set is open, and so its complement $\pi_Y(C)$ is closed. $\square$

8.3. Corollary. *If X is compact then $\pi_Y: X \times Y \rightarrow Y$ is proper.*

PROOF. This follows immediately from Theorem 7.13 and Proposition 8.2. $\square$

8.4. Corollary. *If X and Y are both compact, then $X \times Y$ is compact.* $\square$

8.5. Corollary (Tychonoff Theorem for Finite Products). *If the X_i are compact then $X_1 \times \cdots \times X_n$ is compact.* $\square$

8.6. Corollary. *The cube $I^n \subset \mathbf{R}^n$ is compact.* $\square$

8.7. Corollary. *A subspace of $\mathbf{R}^n$ is compact $\Leftrightarrow$ it is closed and bounded.*

PROOF. Let X be the subspace in question.

($\Rightarrow$) Since X is compact, it is closed. Cover X by the open balls of radius k about the origin, $k = 1, 2, \dots$. Since this has, by hypothesis, a finite subcover, X must be in one of these balls, and hence is bounded.

($\Leftarrow$) If X is closed and bounded, then it is in some ball of radius k about the origin, which in turn is contained in $[-k, k] \times \cdots \times [-k, k]$ (n times), which is compact. Thus X is a closed subset of a compact set and so is compact by Theorem 7.7. $\square$

8.8. Proposition. *A net in a product space $X = \prod X_\alpha$ converges to the point $(\dots, x_\alpha, \dots) \Leftrightarrow$ its composition with each projection $\pi_\alpha: X \rightarrow X_\alpha$ converges to x_α .*

PROOF. This is an easy exercise in the definition of product spaces and of convergence of nets, which will be left to the reader. $\square$

8.9. Theorem (Tychonoff). *The product of an arbitrary collection of compact spaces is compact.*

PROOF. Let $X = \prod X_\alpha$ where the X_α are compact. Let $f: D \rightarrow X$ be a universal net in X . Then the composition $\pi_\alpha \circ f$ is also a universal net by Proposition 6.12. Therefore this composition converges, say to x_α by Theorem 7.14. But this means that the original net converges to the point whose α th coordinate is x_α by Proposition 8.8 and so X is compact by Theorem 7.14. $\square$

Tychonoff's Theorem has the reputation of being difficult. So, how can we prove it with such ease here? The answer is that the entire difficulty has been subsumed in the results about universal nets. The basic facts about universal nets depend on the axiom of choice, and so it follows that so does the Tychonoff Theorem. In fact, it is known that the Tychonoff Theorem is equivalent to the axiom of choice. That is why we gave a separate treatment of the finite case, which does not depend on the axiom of choice. (Also, the finite case is all that is needed in the main body of this book.)

If X is a space and A is a set, the product of A copies of X is often denoted by X^A and can be thought of as the space of functions $f: A \rightarrow X$. In this context, Proposition 8.8 takes the following form:

8.10. Proposition. *A net $\{f_\alpha\}$ in X^A converges to $f \in X^A \Leftrightarrow \forall x \in X, f_\alpha(x) \rightarrow f(x)$. In particular, $\lim(f_\alpha(x)) = (\lim f_\alpha)(x)$.* $\square$

When A also has a topology, the notation X^A is often used for the set of all *continuous* functions $f: A \rightarrow X$. In that context a topology is often used on this set that differs from the product topology. There are several useful topologies in particular circumstances, and so the context must indicate what topology, if any, is meant by this notation.

8.11. Definition. If X and Y are spaces, then their *topological sum* or *disjoint union* $X + Y$ is the set $X \times \{0\} \cup Y \times \{1\}$ with the topology making $X \times \{0\}$ and $Y \times \{1\}$ clopen and the inclusions $x \mapsto (x, 0)$ of $X \rightarrow X + Y$ and $y \mapsto (y, 1)$ of $Y \rightarrow X + Y$ homeomorphisms to their images. More generally, if $\{X_\alpha | \alpha \in A\}$ is an indexed family of spaces then their *topological sum* $\bigoplus_\alpha X_\alpha$ is $\bigcup \{X_\alpha \times \{\alpha\} | \alpha \in A\}$ given the topology making each $X_\alpha \times \{\alpha\}$ clopen and each inclusion $x \mapsto (x, \beta)$ of $X_\beta \rightarrow \bigoplus_\alpha X_\alpha$ a homeomorphism to its image $X_\beta \times \{\beta\}$.

In ordinary parlance, if X and Y are disjoint spaces, one regards $X + Y$ as $X \cup Y$ with the topology making X and Y open subspaces.

PROBLEMS

1. Let X and Y be metric spaces. Define a metric on $X \times Y$ by

$$\text{dist}(\langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle) = (\text{dist}(x_1, x_2)^2 + \text{dist}(y_1, y_2)^2)^{1/2}.$$

Show that the topology induced by this metric is the product topology.

2. Do the same as Problem 1 for the metric:

$$\text{dist}(\langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle) = \max\{\text{dist}(x_1, x_2), \text{dist}(y_1, y_2)\}.$$

3. $\diamond$ For a collection of spaces Y_α show that a function $f: X \rightarrow \prod \{Y_\alpha\}$ is continuous $\Leftrightarrow$ each composition $X \rightarrow \prod \{Y_\alpha\} \rightarrow Y_\alpha$, with the projection, is continuous.
4. $\diamond$ Show that an arbitrary product of Hausdorff spaces is Hausdorff. Also show that an arbitrary product of regular spaces is regular. (*Hint:* Use Proposition 5.2 for the latter.)
5. $\diamond$ If X is a topological space, the “diagonal” of $X \times X$ is the subspace $\Delta = \{\langle x, x \rangle | x \in X\}$. Show that X is Hausdorff $\Leftrightarrow \Delta$ is closed in $X \times X$.
6. $\diamond$ Let $f, g: X \rightarrow Y$ be two maps. If Y is Hausdorff then show that the subspace $A = \{x \in X | f(x) = g(x)\}$ is closed in X .
7. Give an alternative proof of Proposition 8.2 using nets.
8. $\blacklozenge$ Let A be an uncountable set. For each $\alpha \in A$ let $X_\alpha = \{0, 1\}$ with the discrete topology. Put $X = \prod_{\alpha \in A} X_\alpha$. (That is, $X = \{0, 1\}^A$.) Let $p \in X$ be the point with all components $p_\alpha = 1$. Let $K = \{q \in X | q_\alpha = 0 \text{ except for a countable number of } \alpha\}$.
 - (a) Show that p does not have a countable neighborhood basis.
 - (b) Show that there is no neighborhood basis for p simply ordered by inclusion.
 - (c) Show that $\bar{K} = X$ but that if H is a countable subset of K then $\bar{H} \subsetneq K$.
 - (d) Give an explicit description of a net in K which converges to p .

9. ♦ ⇐ Show that a product of a family of connected spaces is connected. Do the same for arcwise connectivity.

9. Metric Spaces Again

In this section we discuss the central concept of “completeness” of a metric space, which says, intuitively, that sequences that should converge do, in fact, converge. We also show that certain topological conditions on a topological space suffice for the existence of a metric on that space consistent with the given topology.

9.1. Definition. A *Cauchy sequence* in a metric space is a sequence $x_1, x_2, x_3, \dots$ such that $\forall \epsilon > 0, \exists N > 0 \ni n, m > N \Rightarrow \text{dist}(x_n, x_m) < \epsilon$.

9.2. Definition. A metric space X is called *complete* if every Cauchy sequence in X converges in X .

9.3. Definition. A metric space X is *totally bounded* if, for each $\epsilon > 0$, X can be covered by a finite number of ϵ -balls.

9.4. Theorem. In a metric space X the following conditions are equivalent:

- (1) X is compact.
- (2) Each sequence in X has a convergent subsequence.
- (3) X is complete and totally bounded.

PROOF. (1) $\Rightarrow$ (2) Let $\{x_n\}$ be a sequence. Suppose that x is not a limit of a subsequence. Then there is an open neighborhood U_x of x containing x_n for only a finite number of n . Since X can be covered by a finite number of the U_x , this contradicts the infinitude of indexes n .

(2) $\Rightarrow$ (3) Let $\{x_n\}$ be a Cauchy sequence. It follows from (2) that some subsequence $x_{n_j} \rightarrow x$ for some $x \in X$. The triangle inequality then implies that $x_n \rightarrow x$ and hence X is complete. Now suppose that X is not covered by a finite number of ϵ -balls. Then one can choose points $x_1, x_2, \dots$ such that $\text{dist}(x_i, x_j) > \epsilon$ for all $j < i$. It follows that the distance between any two of these points is greater than ϵ . Such a sequence can have no convergent subsequences, contrary to (2). So, in fact, X must be totally bounded.

(3) $\Rightarrow$ (2) Let $\{x_n\}$ be an arbitrary sequence in X . Since X is totally bounded by assumption, it can be covered by a finite number of 1-balls. Thus some one of these 1-balls, say B_1 , must contain x_n for an infinite number of n . Next, B_1 , and hence X , can be covered by a finite number of $\frac{1}{2}$ -balls and so one of these balls, say B_2 , must be such that $B_1 \cap B_2$ contains x_n for an infinite number of n . Continuing in this way we can find, for $n = 1, 2, 3, \dots$, a $(1/n)$ -ball B_n such that $B_1 \cap B_2 \cap \dots \cap B_n$ contains x_i for an infinite number of i . Thus we can choose a subsequence $\{x_{n_i}\}$ such that $x_{n_i} \in B_1 \cap \dots \cap B_i$ for all i . If $i < j$

then it follows that x_{n_i} and x_{n_j} are both in B_i and hence $\text{dist}(x_{n_i}, x_{n_j}) < 1/i$. This implies that this subsequence is Cauchy and so it must converge by completeness.

(2) $\Rightarrow$ (1) Suppose $\{U_\alpha | \alpha \in A\}$ is an open covering of X . Since X is totally bounded (by (2) $\Rightarrow$ (3)), we can find a dense sequence of points $x_1, x_2, \dots$ in X . For each x_i there is a positive integer n such that $B_{1/n}(x_i) \subset U_\alpha$ for some α . Denote one such U_α by $V_{n,i}$. Now, given $x \in X$, there is an n such that $B_{2/n}(x) \subset U_\alpha$ for some α . By density, there is also an i such that $\text{dist}(x_i, x) < 1/n$. Then $B_{1/n}(x_i) \subset B_{2/n}(x) \subset U_\alpha$ so that $V_{n,i}$ is defined. Thus, $x \in B_{1/n}(x_i) \subset V_{n,i}$. Therefore the $V_{n,i}$ cover X and this is a countable subcover of the original cover. Let us rename this countable subcover $\{V_1, V_2, \dots\}$. If this has a finite subcover then we are done. If not then the closed sets

$$\begin{aligned} C_1 &= X - V_1, \\ C_2 &= X - (V_1 \cup V_2), \\ C_3 &= X - (V_1 \cup V_2 \cup V_3), \\ &\dots \end{aligned}$$

are all nonempty. Also note that $C_1 \supset C_2 \supset C_3 \supset \dots$. Choose $x_i \in C_i$ for each i . By our assumption, there is a convergent subsequence $x_{n_i} \rightarrow x$, say. Since $x_{n_i} \in C_n$ for all $n_i > n$, and C_n is closed, x must be in C_n for all n . Thus

$$x \in \bigcap \{C_n\} = X - \bigcup \{V_n\} = \emptyset.$$

This contradiction completes the proof. $\square$

It clearly would be desirable to know when a given topological space can be given the structure of a metric space, in which case the space is called "metrizable." There are several known theorems of this nature. We shall be content with giving one of the simpler criteria. This development will span the rest of this section.

9.5. Definition. A Hausdorff space X is said to be *completely regular*, or $T_{3\frac{1}{2}}$, if, for each point $x \in X$ and closed set $C \subset X$ with $x \notin C$, there is a map $f: X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f \equiv 1$ on C .

By following such a function with a map $[0, 1] \rightarrow [0, 1]$ which is 0 on $[0, \frac{1}{2}]$ and stretches $[\frac{1}{2}, 1]$ onto $[0, 1]$, we see that the function f in Definition 9.5 can be taken so that it is 0 on a neighborhood of x .

9.6. Proposition. Suppose X is a metric space. Define:

$$\text{dist}'(x, y) = \begin{cases} 1 & \text{if } \text{dist}(x, y) > 1, \\ \text{dist}(x, y) & \text{if } \text{dist}(x, y) \leq 1. \end{cases}$$

Then dist and dist' give rise to the same topology on X .

PROOF. It is clear that the topology only depends on the open ϵ -balls for small ϵ , and these are the same in the two metrics. $\square$

9.7. Proposition. Let $X_i, i = 1, 2, 3, \dots$, be a metric space with metric bounded by 1 (see Proposition 9.6). Define a metric on $\times \{X_i\}$ by $\text{dist}(x, y) = \sum_i \text{dist}(x_i, y_i)/2^i$, where x_i is the i th coordinate of x , etc. Then this metric gives rise to the product topology.

PROOF. Let X denote the product space with the product topology, and X' the same set with the metric topology. By Problem 3 of Section 8, to show that $X' \rightarrow X$ is continuous, it suffices to show that its composition with the projection to each X_i is continuous. But this projection decreases distance and then multiplies it by the constant 2^i and that clearly implies continuity. For the converse, it suffices to show that for any point $x \in X$, the ϵ -ball about x contains a neighborhood of x in the product topology. Recall that

$$B_\epsilon(x) = \left\{ y \mid \sum_i (\text{dist}(x_i, y_i)/2^i) < \epsilon \right\}.$$

Let n be so large that $2^{-n} < \epsilon/4$ and then let $y_i \in X_i$ be such that $\text{dist}(x_i, y_i) < \epsilon/2$ for $i = 1, 2, \dots, n-1$ and arbitrary for $i \geq n$. Then we compute

$$\begin{aligned} \text{dist}(x, y) &= \sum_{i=1}^{n-1} \text{dist}(x_i, y_i)/2^i + \sum_{i=n}^{\infty} \text{dist}(x_i, y_i)/2^i \\ &< \sum_{i=1}^{n-1} \epsilon/2^{i+1} + \frac{1}{4}\epsilon \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots \right) \\ &< \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

Thus

$$x \in B_{\epsilon/2}(x_1) \times \dots \times B_{\epsilon/2}(x_{n-1}) \times X_n \times X_{n+1} \times \dots \subset B_\epsilon(x)$$

and the middle term is a basic open set in the product topology. $\square$

9.8. Lemma. Suppose that X is Hausdorff and that $f_i: X \rightarrow [0, 1]$ are maps ($i = 1, 2, 3, \dots$) such that, for any point $x \in X$ and any closed set $C \subset X$ with $x \notin C$, there is an index i such that $f_i(x) = 0$ and $f_i \equiv 1$ on C . Define $f: X \rightarrow \times \{[0, 1] \mid i = 1, 2, 3, \dots\}$ by $f(x) = \times \{f_i(x) \mid i = 1, 2, 3, \dots\}$. Then f is an embedding, i.e., a homeomorphism onto its image.

PROOF. f is continuous by Problem 3 of Section 8. It is also clear that f is one-one (but not onto). Thus it suffices to show that: $C \subset X$ closed $\Rightarrow f(C)$ is closed in $f(X)$. Suppose we have a sequence $c_i \in C$ such that $f(c_i) \rightarrow f(x)$. It then suffices to show that $x \in C$. If not, then there is an index i such that $f_i(x) = 0$ and $f_i \equiv 1$ on C . Then $1 = f_i(c_n) \rightarrow f_i(x) = 0$ and this contradiction concludes the proof. $\square$

9.9. Lemma. Suppose that X is a second countable and completely regular space and let $\mathbf{S}$ be a countable basis for the open sets. For each pair $U, V \in \mathbf{S}$ with $\bar{U} \subset V$, select a map $f: X \rightarrow [0, 1]$ which is 0 on U and 1 on $X - V$, provided such a function exists. Call this set of maps $\mathbf{F}$, possibly empty, and note that

F is countable. Then for each $x \in X$ and each closed set $C \subset X$ with $x \notin C$, there is an $f \in \mathbf{F}$ with $f \equiv 0$ on a neighborhood of x and $f \equiv 1$ on C .

PROOF. The whole point of the lemma is, of course, that the map f can be chosen from the previously defined countable collection **F**. Given $x \notin C$ as stated, we can find a $V \in \mathbf{S}$ with $x \in V \subset X - C$ (by definition of a basis). Since X is completely regular we can find a map $g: X \rightarrow [0, 1]$ which is 0 at x and 1 on $X - V$. As remarked below Definition 9.5, this can be assumed to be 0 on a neighborhood of x . This contains a neighborhood $U \in \mathbf{S}$ and so we have provided a triple U, V, g satisfying the initial requirements in the lemma. By assumption, this g can be replaced by another map $f \in \mathbf{F}$ with the same properties and this f clearly satisfies the final requirements. $\square$

9.10. Theorem (Urysohn Metrization Theorem). *If a space X is second countable and completely regular then it is metrizable.*

PROOF. Find a countable family **F** of functions satisfying Lemma 9.9. Apply Lemma 9.8 to obtain an embedding of X into a countable (!) product of unit intervals. Finally, apply Proposition 9.7 to see that this countable product of intervals, and hence X , is metrizable. $\square$

The following lemma will be useful to us later on in the book. The diameter, $\text{diam}(A)$, of a subset A of a metric space is $\sup\{\text{dist}(p, q) \mid p, q \in A\}$.

9.11. Lemma (Lebesgue Lemma). *Let X be a compact metric space and let $\{U_\alpha\}$ be an open covering of X . Then there is a $\delta > 0$ (a “Lebesgue number” for the covering) such that $(A \subset X, \text{diam}(A) < \delta) \Rightarrow A \subset U_\alpha$ for some α .*

PROOF. For each $x \in X$ there is an $\epsilon(x) > 0$ such that $B_{2\epsilon(x)}(x) \subset U_\alpha$ for some α . Then X is covered by a finite number of the balls $B_{\epsilon(x)}(x)$, say for $x = x_1, \dots, x_n$. Define $\delta = \min\{\epsilon(x_i) \mid i = 1, \dots, n\}$. Suppose $\text{diam}(A) < \delta$ and pick a point $a_0 \in A$. Then there is an index $1 \leq i \leq n$ such that $\text{dist}(a_0, x_i) < \epsilon(x_i)$. If $a \in A$, then $\text{dist}(a, a_0) < \delta \leq \epsilon(x_i)$. By the triangle inequality, $\text{dist}(a, x_i) < 2\epsilon(x_i)$. Thus $A \subset B_{2\epsilon(x_i)}(x_i) \subset U_\alpha$ for some α . $\square$

PROBLEMS

1. Show that a countable product of copies of the real line is metrizable.
2. $\diamond$ Show that a subspace of a completely regular space is completely regular.
3. Let X be a metric space. If $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences in X such that $\text{dist}(x_n, y_n) \rightarrow 0$ then call $\{x_n\}$ and $\{y_n\}$ “equivalent.” Let Y be the set of equivalence classes $[\{x_n\}]$ of Cauchy sequences $\{x_n\}$ in X . Give Y the metric

$$\text{dist}([\{x_n\}], [\{y_n\}]) = \lim \text{dist}(x_n, y_n).$$

- (a) Show that this is a metric on Y .

- (b) Show that the function $f: X \rightarrow Y$ given by $x \mapsto [\{x\}]$ is an isometric embedding of X as a dense subspace of Y . ("Isometric" means "preserving distance.")
 - (c) Show that Y is complete. (It is called the "completion" of X .)
 - (d) If $g: X \rightarrow Z$ is an isometry (into) and Z is complete then show that there is a unique factorization $X \xrightarrow{f} Y \xrightarrow{h} Z$ of g with h an isometry.
 - (e) If $g(X)$, in part (d), is dense in Z then show that h is onto.
4. Show that a completely regular space is regular.
5. ♦ Show that an uncountable product of unit intervals is not first countable and hence is not metrizable.

10. Existence of Real Valued Functions

In the metrization theorem of the last section, we gave conditions for metrizability that included complete regularity of the space. This relies on knowing about the existence of sufficiently many, in some sense, continuous real valued functions on the space. That leaves open the question of finding purely *topological* assumptions that will guarantee such functions, and that is what we are going to address in this section.

10.1. Lemma. *Suppose that, on a topological space X , we are given, for each dyadic rational number $r = m/2^n$ ($0 \leq m \leq 2^n$), an open set U_r such that $r < s \Rightarrow \bar{U}_r \subset U_s$. Then the function $f: X \rightarrow \mathbf{R}$ defined by*

$$f(x) = \begin{cases} \inf\{r \mid x \in U_r\} & \text{if } x \in U_1, \\ 1 & \text{if } x \notin U_1, \end{cases}$$

is continuous.

PROOF. Note that, for r dyadic:

$$\begin{aligned} f(x) < r &\Rightarrow x \in U_r & \text{hence} & & f(x) \geq r &\Leftarrow x \notin U_r, \\ f(x) \leq r &\Leftarrow x \in U_r & \text{hence} & & f(x) > r &\Rightarrow x \notin U_r \Rightarrow x \in X - U_r. \end{aligned}$$

Thus, for α real,

$$f^{-1}(-\infty, \alpha) = \{x \mid f(x) < \alpha\} = \bigcup \{U_r \mid r < \alpha\}$$

which is open, and

$$f^{-1}(\beta, \infty) = \{x \mid f(x) > \beta\} = \bigcup \{X - U_r \mid r > \beta\} = \bigcup \{X - \bar{U}_s \mid s > \beta\}$$

which is also open. Since these half infinite intervals give a subbasis for the topology of $\mathbf{R}$, f is continuous. (See Problem 4 of Section 2.) $\square$

10.2. Lemma (Urysohn's Lemma). *If X is normal and $F \subset U$ where F is closed and U is open, then there is a map $f: X \rightarrow [0, 1]$ which is 0 on F and 1 on $X - U$.*

PROOF. Put $U_1 = U$ and use normality to find $F \subset U_0$, $\bar{U}_0 \subset U_1$,

$$\bar{U}_0 \subset U_{1/2} \quad \text{and} \quad \bar{U}_{1/2} \subset U_1,$$

$$\bar{U}_0 \subset U_{1/4} \quad \text{and} \quad \bar{U}_{1/4} \subset U_{1/2} \quad \text{and} \quad \bar{U}_{1/2} \subset U_{3/4} \quad \text{and} \quad \bar{U}_{3/4} \subset U_1,$$

and so on. Apply Lemma 10.1. $\square$

10.3. Corollary. *Normality $\Rightarrow$ Complete Regularity.* $\square$

10.4. Theorem (Tietze Extension Theorem). *Let X be normal and $F \subset X$ be closed and let $f: F \rightarrow \mathbf{R}$ be continuous. Then there is a map $g: X \rightarrow \mathbf{R}$ such that $g(x) = f(x)$ for all $x \in F$. Moreover, it can be arranged that*

$$\sup_{x \in F} f(x) = \sup_{x \in X} g(x) \quad \text{and} \quad \inf_{x \in F} f(x) = \inf_{x \in X} g(x).$$

PROOF. First let us take the case in which f is bounded. Without loss of generality, we can assume $0 \leq f(x) \leq 1$ with infimum 0 and supremum 1. By the Urysohn Lemma (Lemma 10.2), there exists a function $g_1: X \rightarrow [0, \frac{1}{3}]$ such that

$$g_1(x) = \begin{cases} 0 & \text{if } x \in F \text{ and } f(x) \leq \frac{1}{3}, \\ \frac{1}{3} & \text{if } x \in F \text{ and } f(x) \geq \frac{2}{3}. \end{cases}$$

Put $f_1 = f - g_1$ and note that $0 \leq f_1(x) \leq \frac{2}{3}$ for all $x \in F$.

Repeating this, find $g_2: X \rightarrow [0, \frac{1}{3} \cdot \frac{2}{3}]$ such that

$$g_2(x) = \begin{cases} 0 & \text{if } x \in F \text{ and } f_1(x) \leq \frac{1}{3} \cdot \frac{2}{3}, \\ \frac{1}{3} \times \frac{2}{3} & \text{if } x \in F \text{ and } f_1(x) \geq \frac{2}{3} \cdot \frac{2}{3}. \end{cases}$$

Put $f_2 = f_1 - g_2$ and note that $0 \leq f_2(x) \leq (\frac{2}{3})^2$ for all $x \in F$.

For the inductive step, suppose we have defined a function f_n with $0 \leq f_n(x) \leq (\frac{2}{3})^n$ for $x \in F$. Then find $g_{n+1}: X \rightarrow [0, (\frac{1}{3})(\frac{2}{3})^n]$ such that

$$g_{n+1}(x) = \begin{cases} 0 & \text{if } x \in F \text{ and } f_n(x) \leq (\frac{1}{3})(\frac{2}{3})^n, \\ (\frac{1}{3})(\frac{2}{3})^n & \text{if } x \in F \text{ and } f_n(x) \geq (\frac{2}{3})(\frac{2}{3})^n. \end{cases}$$

Put $f_{n+1} = f_n - g_{n+1}$.

Now put $g(x) = \sum g_n(x)$. This series converges uniformly since $0 \leq g_n(x) \leq (\frac{1}{3})(\frac{2}{3})^{n-1}$. Thus g is continuous, by Theorem 2.12.

For $x \in F$ we have

$$\begin{aligned} f - g_1 &= f_1, \\ f_1 - g_2 &= f_2, \\ &\dots \end{aligned}$$

By adding and canceling we get

$$f - (g_1 + g_2 + \dots + g_n) = f_n \quad \text{and} \quad 0 \leq f_n(x) \leq (\frac{2}{3})^n,$$

and taking the limit gives that $g(x) = f(x)$ on F . Clearly the bounds are also correct.

Now we consider the unbounded cases:

Case I: f is unbounded in both directions.

Case II: f is bounded below by a .

Case III: f is bounded above by b .

Let h be a homeomorphism:

$$\begin{aligned} (-\infty, \infty) &\rightarrow (0, 1) && \text{in Case I,} \\ [a, \infty) &\rightarrow [0, 1) && \text{in Case II,} \\ (-\infty, b] &\rightarrow (0, 1] && \text{in Case III.} \end{aligned}$$

Then $h \circ f$ is bounded by 0, 1 and we can extend it to g_1 say. If we can arrange that $g_1(x)$ is never 0 (resp. 1) if $h \circ f$ is never 0 (resp. 1) then $g = h^{-1} \circ g_1$ would be defined and would extend f .

Thus put

$$\begin{aligned} C &= \{x | g_1(x) = 0 \text{ or } 1\} && \text{in Case I,} \\ C &= \{x | g_1(x) = 1\} && \text{in Case II,} \\ C &= \{x | g_1(x) = 0\} && \text{in Case III.} \end{aligned}$$

Then C is closed and $C \cap F = \emptyset$, so there exists a function $k: X \rightarrow [0, 1]$ such that $k \equiv 0$ on C and $k \equiv 1$ on F . Put $g_2 = k \cdot g_1 + (1 - k) \cdot \frac{1}{2}$. Then g_2 is always between g_1 and $\frac{1}{2}$ with $g_2 \neq g_1$ on C . Also, $g_2 = g_1 = h \circ f$ on F . Thus $g = h^{-1} \circ g_2$ extends f in the desired manner. $\square$

PROBLEMS

1. If X is a compact Hausdorff space then show that its quasi-components are connected (and hence that its quasi-components coincide with its components). [Hint: If C is a quasi-component, let $C = \bigcap C_\alpha$ where the C_α are the clopen sets containing C . If C is disconnected, then $C = A \cup B$, $A \cap B = \emptyset$, A, B closed. Let $f: X \rightarrow [0, 1]$ be 0 on A and 1 on B . Put $U = f^{-1}([0, \frac{1}{2}))$ and apply Problem 2 of Section 7.]
2. $\diamond$ If F is a closed subspace of the normal space X then show that any map $F \rightarrow \mathbf{R}^n$ can be extended to X .

11. Locally Compact Spaces

There are many spaces, the most important being euclidean spaces, which are not compact but which contain enough compact subspaces to be important for many properties of the space itself. One class of such spaces is the subject of this section.

11.1. Definition. A topological space is said to be *locally compact* if every point has a compact neighborhood.

11.2. Theorem. *If X is a locally compact Hausdorff space then each neighborhood of a point $x \in X$ contains a compact neighborhood of x . (That is, the compact neighborhoods of x form a neighborhood basis at x .) In particular, X is completely regular.*

PROOF. Let C be a compact neighborhood of x and U an arbitrary neighborhood of x . Let $V \subset C \cap U$ be open with $x \in V$. Then $\bar{V} \subset C$ is compact Hausdorff and therefore regular. Thus there exists a neighborhood $N \subset V$ of x in C which is closed in $\bar{V}$ and hence closed in X . Since N is closed in the compact space C , it is compact by Theorem 7.7. Since N is a neighborhood of x in $\bar{V}$ and since $N = N \cap V$, N is a neighborhood of x in the open set V and hence in X . $\square$

11.3. Theorem. *Let X be a locally compact Hausdorff space. Put $X^+ = X \cup \{\infty\}$ where ∞ just represents some point not in X . Define an open set in X^+ to be either an open set in $X \subset X^+$ or $X^+ - C$ where $C \subset X$ is compact. Then this defines a topology on X^+ which makes X^+ into a compact Hausdorff space called the "one-point compactification" of X . Moreover, this topology on X^+ is the only topology making X^+ a compact Hausdorff space with X as a subspace.*

PROOF. The whole space X^+ and $\emptyset$ are clearly open. If $V \subset X$ is open and $U = X^+ - C$ with $C \subset X$ compact then $U \cap V = V - C$ which is open in X (C being closed in X by Theorem 7.5). The other cases of an intersection of two open sets are trivial.

For arbitrary unions of open sets, let $U = \bigcup \{U_\alpha\}$. If all the U_α are open subsets of X then the union is clearly open. If some $U_\beta = X^+ - C$ then $X^+ - U = \bigcap \{X^+ - U_\alpha\} = C \cap (\bigcap \{X - U_\alpha \mid \alpha \neq \beta\})$ which is closed in C and therefore compact. Thus, this is a topology.

Suppose that $\{U_\alpha\}$ is an open cover of X^+ . One of these sets, say U_β , contains $\{\infty\}$. Then $X - U_\beta$ is compact and hence is covered by a finite subcollection of the other U_α . Therefore X^+ is compact.

To see that X^+ is Hausdorff, it clearly suffices to separate ∞ from any point $x \in X$. Let V be an open neighborhood of x in X such that $\bar{V} \subset X$ is compact. Then $x \in V$ and $\infty \in X - \bar{V}$ provide the required separation.

For uniqueness, let $U \subset X^+$ be an open set in some such topology. Then $C = X^+ - U$ is closed and therefore compact. If $C \subset X$ then U is open in the described topology. If $C \not\subset X$ then $U \subset X$ and must be open in X since X is a subspace. Thus, again U is open in the described topology. It remains to show that we are forced to take the described open sets as open. Since X is a subspace, if $U \subset X$ is open in X then $U = U' \cap X$ for some U' open in X^+ . But X is an open subset of X^+ since points are closed in a Hausdorff space, so $U = U' \cap X$ is open in X^+ . Next, if C is compact in X then it is compact in X^+ , since compactness does not depend on the containing space, and thus C is closed in X^+ . It follows that $X^+ - C$ is open in X^+ . $\square$

Note that if X is already compact, then ∞ is an isolated point (clopen) in X^+ and X is also clopen in X^+ .

11.4. Theorem. *Suppose that X and Y are locally compact, Hausdorff spaces and that $f: X \rightarrow Y$ is continuous. Then f is proper $\Leftrightarrow f$ extends to a continuous $f^+: X^+ \rightarrow Y^+$ by setting $f^+(\infty_X) = \infty_Y$.*

PROOF. $\Rightarrow$: f^+ exists as a function, so it suffices to check continuity of it. Suppose $U \subset Y^+$ is open. In case $U \subset Y$ then $(f^+)^{-1}(U) = f^{-1}(U)$ is open. In the other case, $U = Y^+ - C$ with $C \subset Y$ compact. Then $(f^+)^{-1}(U) = X^+ - f^{-1}(C)$ is open in X^+ since $f^{-1}(C)$ is compact, and therefore closed, by properness.

$\Leftarrow$: If f^+ exists then $(f^+)^{-1}(\infty_Y) = \{\infty_X\}$ and thus $(f^+)^{-1}(Y) = X$. If $C \subset Y$ is compact then it is closed and so $f^{-1}(C)$ is closed in X^+ and hence is compact and is contained in X . Thus f is proper. $\square$

11.5. Proposition. *If $f: X \rightarrow Y$ is a proper map between locally compact Hausdorff spaces, then f is closed.*

PROOF. There is an extension $f^+: X^+ \rightarrow Y^+$. If $F \subset X$ is closed in X then $F \cup \{\infty\}$ is closed in X^+ and hence compact. Consequently, $f^+(F \cup \{\infty\})$ is compact by Theorem 7.6 and hence closed in Y^+ by Theorem 7.5. But then $f(F) = f^+(F \cup \{\infty\}) \cap Y$ is closed in Y . $\square$

11.6. Definition. A subspace A of a topological space is said to be *locally closed* if each point $a \in A$ has an open neighborhood U_a such that $U_a \cap A$ is closed in U_a .

11.7. Proposition. *A subspace $A \subset X$ is locally closed $\Leftrightarrow$ it has the form $A = C \cap U$ where U is open in X and C is closed in X .*

PROOF. Put $U = \bigcup \{U_a | a \in A\}$, as in Definition 11.6, which is open, and $C = \bar{A}$ which is closed. Then

$$C \cap U = \bar{A} \cap \left(\bigcup U_a \right) = \bigcup (\bar{A} \cap U_a) = \bigcup (A \cap U_a) = A \cap U = A. \quad \square$$

11.8. Theorem. *For a Hausdorff space X the following conditions are equivalent:*

- (1) X is locally compact.
- (2) X is a locally closed subspace of a compact Hausdorff space.
- (3) X is a locally closed subspace of a locally compact Hausdorff space.

PROOF. If X is locally compact then it is an open subspace of its one-point compactification. Thus (1) $\Rightarrow$ (2). Clearly, (2) $\Rightarrow$ (3). If $Y \supset X$ is locally compact and $X = C \cap U$ where $C \subset Y$ is closed and $U \subset Y$ is open, then C is locally

compact, and $X = U \cap C$ is open in C and hence is also locally compact. Thus (3) $\Rightarrow$ (1). $\square$

The remainder of this section is not used in the remainder of this book and so can be skipped. It assumes knowledge of nets from Section 6.

The preceding results suggest the question of when a topological space X can be embedded in a compact, Hausdorff space Y (as a subspace). Since Y is normal, it is also completely regular. Since a subspace of a completely regular space is completely regular, it follows that X must be completely regular. This turns out to be the precise condition needed.

If X is a completely regular space, consider the set $\mathbf{F}$ of all maps $f: X \rightarrow [0, 1]$. Define

$$\Phi: X \rightarrow [0, 1]^{\mathbf{F}} = \times \{[0, 1] \mid f \in \mathbf{F}\}$$

by $\Phi(x)(f) = f(x)$. (Here we regard an element of $[0, 1]^{\mathbf{F}}$ as a function $\mathbf{F} \rightarrow [0, 1]$.)

11.9. Definition. If X is a completely regular space, and $\Phi: X \rightarrow [0, 1]^{\mathbf{F}}$ is defined as above then the closure of $\Phi(X)$ is called the *Stone-Čech compactification* of X and is denoted by $\beta(X)$.

11.10. Theorem. If X is a completely regular space, then $\beta(X)$ is compact Hausdorff and $\Phi: X \rightarrow \beta(X)$ is an embedding.

PROOF. The function Φ is one-one since, if $\Phi(x) = \Phi(y)$, then $f(x) = f(y)$ for all maps $f: X \rightarrow [0, 1]$ and this implies that $x = y$ by complete regularity.

To prove continuity, let x_α be a net in X converging to x . Then

$$\lim (\Phi(x_\alpha)(f)) = \lim (f(x_\alpha)) = f(x) = \Phi(x)(f)$$

for all maps $f: X \rightarrow [0, 1]$. This implies that $\lim \Phi(x_\alpha) = \Phi(x)$ by Proposition 8.8.

For continuity of the inverse, suppose that $\{x_\alpha\}$ is a net in X such that $\Phi(x_\alpha)$ converges to $\Phi(x)$. Then, for all maps $f: X \rightarrow [0, 1]$,

$$\lim (f(x_\alpha)) = \lim (\Phi(x_\alpha)(f)) = \Phi(x)(f) = f(x).$$

If x_α does not converge to x then there is a neighborhood U of x such that x_α is frequently in $X - U$. But there is a map $f: X \rightarrow [0, 1]$ which is 0 at x and 1 on $X - U$. Thus $f(x_\alpha)$ is frequently 1, while $f(x) = 0$ contradicting the convergence of $f(x_\alpha)$ to $f(x)$. $\square$

11.11. Theorem. If X is completely regular and $f: X \rightarrow \mathbf{R}$ is a bounded real valued map, then f can be extended uniquely to a map $\beta(X) \rightarrow \mathbf{R}$.

PROOF. It suffices to treat the case in which the image of f is in $[0, 1]$. Consider the function $\bar{f}: [0, 1]^{\mathbf{F}} \rightarrow \mathbf{R}$ defined by $\bar{f}(\mu) = \mu(f)$. If $\{\mu_\alpha\}$ is a net in

$[0, 1]^F$ converging to μ then

$$\lim(\bar{f}(\mu_x)) = \lim(\mu_x(f)) = \mu(f) = \bar{f}(\mu),$$

which shows that $\bar{f}$ is continuous.

If $x \in X$ then $\bar{f}(\Phi(x)) = (\Phi(x))(f) = f(x)$, showing that $\bar{f}$ does extend f . $\square$

The problems give other properties of the Stone–Čech compactification.

PROBLEMS

1. Show that the Stone–Čech compactification $\beta(\cdot)$ is a functor on completely regular spaces by showing that a map $f: X \rightarrow Y$ induces a unique commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ \beta(X) & \xrightarrow{\beta(f)} & \beta(Y) \end{array}$$

such that $\beta(f \circ g) = \beta(f) \circ \beta(g)$ and $\beta(1_X) = 1_{\beta(X)}$.

2. Show that the Stone–Čech compactification is the “largest” compactification of a completely regular space X by showing that if $g: X \hookrightarrow Y$ is any compactification, then there is a unique map $\beta(X) \rightarrow Y$ factoring g .
3. Let ω be the set of natural integers and let X be its Stone–Čech compactification. Show that the sequence given by the usual ordering of ω can have no convergent subsequence in X . Conclude that X is not metrizable and not second countable. (Note that this sequence does have a convergent subnet since that is always true in a compact space.)

12. Paracompact Spaces

The notion of “paracompactness” of a space is a type of localization of compactness. It is very different, however, from local compactness. Paracompact spaces are very close to being metrizable, but the concept of paracompactness is sometimes simpler to deal with than that of a metric. In this book, most spaces in which we shall be interested are paracompact. The most important property of paracompact spaces is the existence of “partitions of unity,” see Definition 12.7.

12.1. Definition. If $\mathbf{U}$ and $\mathbf{V}$ are open coverings of a space then $\mathbf{U}$ is said to be a *refinement* of $\mathbf{V}$ if each element of $\mathbf{U}$ is a subset of some element of $\mathbf{V}$.

12.2. Definition. A collection $\mathbf{U}$ of subsets of a topological space X is said to be *locally finite* if each point $x \in X$ has a neighborhood N which meets, nontrivially, only a finite number of the members of $\mathbf{U}$.

12.3. Definition. A Hausdorff space X is said to be *paracompact* if every open covering of X has an open, locally finite refinement.

12.4. Proposition. *A closed subspace of a paracompact space is paracompact.*

PROOF. If A is a closed subspace of the paracompact space X , cover A with sets open in X . Throw in the set $X - A$. Take a locally finite refinement of this open covering of X and intersect it with A . This gives a locally finite refinement of the original covering of A . $\square$

12.5. Theorem. *A paracompact space is normal.*

PROOF. We will first show the paracompact space X to be regular. Thus suppose $x \in X$ and $C \subset X$ is closed with $x \notin C$. For each point $y \in C$ there are disjoint open sets U_y, V_y with $x \in U_y$ and $y \in V_y$. Cover X by $X - C$ together with the sets V_y . Then there is an open locally finite refinement, say by sets U_α . Let $U = \bigcup \{U_\alpha \mid U_\alpha \subset \text{some } V_y\}$ and note that this contains C . Since this is a locally finite collection, its closure $\bar{U}$ is the union of the closures of the same U_α 's. But x is not in any of the $\bar{U}_\alpha$ and so $x \notin \bar{U}$. Thus U and $X - \bar{U}$ provide the required separation.

The same argument, with C playing the role of x and the other closed set playing the role of C , shows X to be normal. $\square$

Thus paracompact spaces are close to being metric spaces because all that is needed is second countability. Also, it is known that metric spaces are paracompact. (This is very hard to prove and we will not attempt it.) However, there are paracompact spaces that are not metrizable. There are also examples of paracompact spaces having subspaces which are not paracompact, and, of course, that cannot happen with metrizable spaces.

Normality implies that a paracompact space has many real valued maps, a property that we will now exploit.

12.6. Definition. If f is a real valued map then the *support* of f is

$$\text{support}(f) = \text{closure} \{x \mid f(x) \neq 0\}.$$

12.7. Definition. Let $\{U_\alpha \mid \alpha \in A\}$ be an open covering of the space X . Then a *partition of unity* subordinate to this covering is a collection of maps

$$\{f_\beta: X \rightarrow [0, 1] \mid \beta \in B\}$$

such that:

- (1) There is a locally finite open refinement $\{V_\beta \mid \beta \in B\}$ such that $\text{support}(f_\beta) \subset V_\beta$ for all $\beta \in B$; and
- (2) $\sum_\beta f_\beta(x) = 1$ for each $x \in X$.

12.8. Theorem. *If X is paracompact and $\mathbf{U}$ is an open covering of X then there exists a partition of unity subordinate to $\mathbf{U}$.*

PROOF. Without loss of generality, we may assume that the covering $\mathbf{U} = \{U_\alpha | \alpha \in A\}$ is locally finite.

Consider a family $F = \{g_\beta: X \rightarrow [0, 1] | \beta \in A_F\}$, where $A_F \subset A$, such that if $\bar{W}_\beta = \{x | g_\beta(x) > 0\}$, then $\bar{W}_\beta \subset U_\beta$ and $\{W_\beta | \beta \in A_F\} \cup \{U_\alpha | \alpha \in A - A_F\}$ covers X . Let $\mathbf{F}$ be a collection of such families F which is simply ordered by inclusion and is a maximal such collection; see Theorem B.18(C). Let $G = \bigcup \mathbf{F}$. We claim that $G \in \mathbf{F}$. Now $G = \{g_\beta | \beta \in A_G = \bigcup A_F \text{ for } F \in \mathbf{F}\}$. If $G \notin \mathbf{F}$ then there is a point $x \in X$ not in any W_β for $\beta \in A_G$ or in any U_α for $\alpha \in A - A_G$. Let $\{\alpha_1, \dots, \alpha_n\}$ be the finite nonempty set of indices for which $x \in U_{\alpha_i}$, all i . Then each $\alpha_i \in A_G$. Since $\mathbf{F}$ is simply ordered there is an $F \in \mathbf{F}$ for which $\alpha_i \in A_F$ for all $i = 1, \dots, n$. But then x must be in some W_{α_i} , a contradiction. Therefore $G \in \mathbf{F}$ as claimed.

Next we claim that $A_G = A$. If not then let $\alpha \in A - A_G$ and put

$$D = \bigcup \{W_\beta | \beta \in A_G\} \cup \bigcup \{U_\gamma | \gamma \in A - A_G, \gamma \neq \alpha\}.$$

Then $X = D \cup U_\alpha$. Let $C = X - D$ which is closed and inside U_α . Since X is normal, there exists an open set V with $C \subset V$ and $\bar{V} \subset U_\alpha$. By Urysohn's Lemma there is a map $g_\alpha: X \rightarrow [0, 1]$ which is 1 on C and 0 outside V . Then $W_\alpha \supset C$ and so $X = D \cup W_\alpha$, showing that $G \cup \{g_\alpha\}$ is a collection of maps, as above, properly containing G . This contradicts the maximality of $\mathbf{F}$, and so $A_G = A$ as claimed.

Thus we now have a collection $\{g_\alpha | \alpha \in A\}$ of maps such that the $W_\alpha = \{x | g_\alpha(x) > 0\}$ cover X and $\bar{W}_\alpha \subset U_\alpha$. Let $g = \sum_\alpha g_\alpha$, which makes sense by the local finiteness, and note that $g(x) > 0$ for all x . Then put $f_\alpha = g_\alpha/g$. This fulfills our requirements. $\square$

Note that we proved a little more than was stated in Theorem 12.8. Namely, if the original covering is already locally finite then it need not be refined as in Definition 12.7. Also note that the sets W_α form a covering and that $\bar{W}_\alpha \subset U_\alpha$. Thus, we have:

12.9. Proposition. *If X is paracompact and $\{U_\alpha\}$ is a locally finite open covering of X then there is an open covering $\{V_\alpha\}$ such that, for each α , $\bar{V}_\alpha \subset U_\alpha$.* $\square$

Generally it is difficult to check that a space is paracompact by just using the definition. Also, we would like to avoid using that metric spaces are paracompact, since we have not proved it. However, the following criterion will apply in most cases of interest to us here.

12.10. Definition. A space is called σ -compact if it is the union of countably many compact subspaces.

12.11. Theorem. *A locally compact, Hausdorff space is paracompact $\Leftrightarrow$ it is the disjoint union of open σ -compact subsets.*

PROOF. $\Rightarrow$: Using local compactness, cover the space with open sets U_α such that $\bar{U}_\alpha$ is compact. Using paracompactness it is easy to see that this covering can be replaced by one which is also locally finite, so assume that. We shall inductively construct open sets V_i whose closures are compact. We start with $V_1 = U_\beta$ for some given β . If V_n has been defined then consider all the U_α which intersect $\bar{V}_n$. By compactness of $\bar{V}_n$ and the local finiteness of the cover, this set of U_α 's is finite. Let V_{n+1} be the union of these U_α . Then $\bar{V}_{n+1}$ is the union of the closures of this finite set of U_α 's and so it is compact. Also $\bar{V}_n \subset V_{n+1}$. Put $V = \bigcup V_n$. Then V is the union of countably many of the U_α 's. By local finiteness $\bar{V}$ is the union of the closures of these U_α 's but each of these closures is contained in some $\bar{V}_n \subset V_{n+1} \subset V$. Thus $\bar{V} = V$ is clopen and, by construction, is σ -compact. The remainder of the proof of the implication ($\Rightarrow$) is accomplished by a Maximality Principle argument which will be left to the reader since this implication will not be used in this book.

$\Leftarrow$: It is clear that a disjoint union of open paracompact spaces is paracompact, and so we may as well assume the space X to be σ -compact; $X = C_1 \cup C_2 \cup \dots$ where the C_i are compact. In sequence, alter the C_i by adding to each C_{i+1} (and to the following ones at the same time) a finite union of compact sets (by local compactness) whose interiors cover C_i . In this way, we get $C_i \subset \text{int}(C_{i+1})$ for all i . Define compact sets $A_1 = C_1$, and $A_i = C_i - \text{int } C_{i-1}$ for $i > 1$. (Think of the C_i as concentric disks and the A_i as the rings between them.) Note that each A_i intersects nontrivially with only (at most) A_{i-1} and A_{i+1} . A little more work, using the compactness, hence normality, of the A_i shows that we can enlarge the A_i slightly to provide compact sets B_i whose interiors contain the A_i and which intersect only with the two B_j with adjoining indices. Now, given an open covering, consider the induced covering of each of the compact sets A_i . We can select a finite refinement still covering A_i and with none of the covering sets overflowing from B_i . It is then clear that these finite coverings taken together provide a locally finite refinement of the original cover. $\square$

12.12. Theorem. *If X is locally compact, Hausdorff, and second countable then its one-point compactification X^+ is metrizable and X is σ -compact and paracompact.*

PROOF. Let $\mathbf{B}$ be a countable basis for X and let $C \subset X$ be compact. If $x \in C$ then x has a compact neighborhood N and there is a member $U_x \in \mathbf{B}$ with $x \in U_x \subset N$. Hence C is covered by a finite union $U_{x_1} \cup \dots \cup U_{x_n}$ of such sets. Put $V = X - \bigcup \bar{U}_{x_i}$. Then $V \cup \{\infty\}$ is a neighborhood of ∞ in X^+ contained in the arbitrary neighborhood $X^+ - C$. These sets V are indexed by finite subsets of $\mathbf{B}$ and hence are countable in number; see the remark below Theorem B.27. This shows that X^+ is second countable, and also shows that

X is σ -compact. By Theorem 12.11, X is paracompact. By Theorem 7.11, Corollary 10.3 and Theorem 9.10, X^+ is metrizable. $\square$

PROBLEMS

1. Without using Theorem 12.11 or the fact that metric spaces are paracompact, show that any open subspace of euclidean space is σ -compact, and hence paracompact by Theorem 12.11.
2. Suppose X is paracompact. For any open subset U of $X \times [0, \infty)$ which contains $X \times \{0\}$ show that there is a map $f: X \rightarrow (0, \infty)$ such that $(x, y) \in U$ for all $y \leq f(x)$.

13. Quotient Spaces

The notion of a quotient space or identification space is of central importance in topology. It gives, for example, a firm foundation for the intuitive idea of the operation of “pasting” spaces together. It also provides many other techniques of producing new spaces out of old ones. It can also be difficult to understand when met for the first time, and the reader is advised to study it fully before going further in this book.

13.1. Definition. Let X be a topological space, Y a set, and $f: X \rightarrow Y$ an onto function. Then we define a topology on Y called the *topology induced by f* or the *quotient topology*, by specifying a set $V \subset Y$ to be open $\Leftrightarrow f^{-1}(V)$ is open in X . Note that this is the largest topology on Y which makes f continuous.

13.2. Definition. Let X be a topological space and $\sim$ an equivalence relation on X . Let $Y = X/\sim$ be the set of equivalence classes and $\pi: X \rightarrow Y$ the canonical map taking $x \in X$ to its equivalence class $[x] \in X/\sim$. Then Y , with the topology induced by π , is called a *quotient space* of X .

Quotient spaces often have very non-Hausdorff topologies. For example, if X is the real line and $x \sim y \Leftrightarrow x - y$ is rational, then $X/\sim$ is an uncountable set but has the trivial topology, as the reader is asked to verify in Problem 6. We will mostly be concerned with quotient spaces that are better behaved.

The reader can verify the following fact directly from the definition:

13.3. Proposition. *A quotient space of a quotient space of X is a quotient space of X . That is, if $X \rightarrow Y \rightarrow Z$ are two onto functions and Y is given the quotient topology from X , and Z is given the quotient topology from Y , then Z has the quotient topology from X induced by the composition of the two functions.* $\square$

13.4. Definition. A map $X \rightarrow Y$ is called an *identification map* if it is onto and Y has the quotient topology.

13.5. Proposition. *A surjection $f: X \rightarrow Y$ is an identification map $\Leftrightarrow$ (for all functions $g: Y \rightarrow Z$, $(g \circ f)$ is continuous $\Leftrightarrow g$ is continuous).*

PROOF. The $\Rightarrow$ part is clear from the definitions. For $\Leftarrow$ specialize to the case $Z = Y$ as sets, with the identification topology (on Z) and g the identity function. Then the composition $X \rightarrow Y \xrightarrow{g} Z$ is continuous, so the condition demands that g be continuous. But g^{-1} is continuous because the composition $X \rightarrow Z \rightarrow Y$ is f which is continuous (by the case $Z = Y$ as spaces and $g = 1_Y$) and since Z has the quotient topology. Thus g is a homeomorphism, meaning that $Y = Z$ as spaces. $\square$

13.6. Example. The projective plane is often defined as the sphere S^2 with antipodal points identified. That is, it is given the quotient topology from the relation that identifies antipodal points in a sphere. A second description of the projective plane one often sees is that it is the unit disk D^2 with antipodal points on the boundary identified. Regard D^2 as the upper hemisphere and consider the diagram:

$$\begin{array}{ccc} D^2 & \xrightarrow{i} & S^2 \\ \downarrow f & & \downarrow g \\ D^2/\sim & \xrightarrow{k} & S^2/\sim \end{array}$$

where the maps f and g are the identifications, i is the inclusion, and k is induced (the only function making the diagram commute). If $U \subset S^2/\sim$ is open then $g^{-1}(U)$ is open so $(gi)^{-1}(U) = i^{-1}(g^{-1}(U))$ is open. But this is the same as $(kf)^{-1}(U) = f^{-1}(k^{-1}(U))$. Thus $k^{-1}(U)$ is open by the definition of the quotient topology. That means that k is continuous. But k is also clearly one-one and onto. Moreover, $D^2/\sim$ is compact since D^2 is. Also, $S^2/\sim$ is easily seen to be Hausdorff, and so we finally conclude that k is a homeomorphism from Theorem 7.8. Thus, indeed, these two ways of defining the projective plane as a topological space are equivalent. This is a typical argument involving spaces obtained via identifications.

An often used special case of quotient spaces is the idea of “collapsing” a subspace:

13.7. Definition. If X is a space and $A \subset X$, then X/A denotes the quotient space obtained via the equivalence relation whose equivalence classes are A and the single point sets $\{x\}$, $x \in X - A$.

The following is an easy exercise:

13.8. Proposition. *If X is regular and A is closed then X/A is Hausdorff. If X is normal and A is closed, then X/A is normal.* $\square$

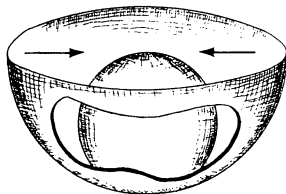


Figure I-1. The sphere as quotient space of a disk.

13.9. Example. Consider the cylinder $S^n \times I$. Define $f: S^n \times I \rightarrow \mathbf{D}^{n+1}$ by $f(x, t) = tx$. This carries the set $S^n \times \{0\}$ to the origin and so f factors through $S^n \times I / S^n \times \{0\}$. The resulting map $g: S^n \times I / S^n \times \{0\} \rightarrow \mathbf{D}^{n+1}$ is clearly one-one and onto. Thus it is a homeomorphism by Theorem 7.8.

13.10. Example. Consider the n -disk $\mathbf{D}^n$. This is clearly homeomorphic to the lower n -hemisphere of radius 2 centered at 1 on the “vertical” axis. (See Figure I-1.) We can map this onto the n -sphere S^n of radius 1 centered at the origin by projection towards the vertical axis. It maps the boundary of the disk to the north pole of the sphere. This function is distance decreasing and hence continuous. Also consider the quotient space $\mathbf{D}^n / S^{n-1}$. One can factor the projection of the disk to the sphere through this space. By an argument similar to that in Example 13.9 one can show that the resulting map $\mathbf{D}^n / S^{n-1} \rightarrow S^n$ is a homeomorphism.

If the method in Example 13.9 is not available, the following gives a criterion for deciding the same sort of question.

13.11. Definition. If $A \subset X$ and if $\sim$ is an equivalence relation on X then the *saturation* of A is $\{x \in X \mid x \sim a \text{ for some } a \in A\}$.

13.12. Proposition. If $A \subset X$ and $\sim$ is an equivalence relation on X such that every equivalence class intersects A nontrivially, then the induced map $k: A/\sim \rightarrow X/\sim$ is a homeomorphism if the saturation of every open (resp. closed) set of A is open (resp. closed) in X .

PROOF. If $f: A \rightarrow A/\sim$ and $g: X \rightarrow X/\sim$ are the canonical maps, and U is an open set in $A/\sim$ then $g^{-1}(k(U))$ is the saturation of $f^{-1}(U)$. Moreover, by definition, it is open $\Leftrightarrow k(U)$ is open in $X/\sim$. Also k is clearly one-one, onto, and continuous. $\square$

Another common application of the idea of a quotient space is a space obtained by “attaching”:

13.13. Definition. Let X and Y be spaces and $A \subset X$ closed. Let $f: A \rightarrow Y$ be a map. Then we denote by $Y \cup_f X$, the quotient space of the disjoint union

$X + Y$ by the equivalence relation $\sim$ which is generated by the relations $a \sim f(a)$ for $a \in A$.

(To be more precise about the equivalence relation, for points u, v in $X + Y$, $u \sim v$ if one of the following is true: (1) $u = v$; (2) $u, v \in A$, and $f(u) = f(v)$; (3) $u \in A$ and $v = f(u) \in Y$.)

Note that if Y is a one-point space then $Y \cup_f X = X/A$.

The following is an easy verification left to the reader:

13.14. Proposition. *The canonical map $Y \rightarrow Y \cup_f X$ is an embedding onto a closed subspace. The canonical map $X - A \rightarrow Y \cup_f X$ is an embedding onto an open subspace.* $\square$

13.15. Definition. If A is a subspace of a space X then a map $f: X \rightarrow A$ such that $f(a) = a$ for all points $a \in A$, is called a *retraction*, and A is said to be a *retract* of X .

Special cases of attachments of importance to us are the “mapping cylinder” and the “mapping cone.” As is usual, the unit interval $[0, 1]$ will be denoted by I here.

13.16. Definition. If $f: X \rightarrow Y$ is a map then the *mapping cylinder* of f is the space $M_f = Y \cup_{f_0} X \times I$ where $f_0: X \times \{0\} \rightarrow Y$ is $f_0(x, 0) = f(x)$. See Figure I-2.

Note that $X \approx X \times \{1\}$ is embedded as a closed subset of M_f . By an abuse of notation, we will regard this as an inclusion $X \subset M_f$. Also note then that there is the factorization of f , $X \subset M_f \xrightarrow{r} Y$ where r is the retraction of M_f onto Y induced by the projection $X \times I \rightarrow X \times \{0\}$.

13.17. Definition. If $f: X \rightarrow Y$ is a map then the *mapping cone* of f is the space $C_f = M_f / (X \times \{1\})$.

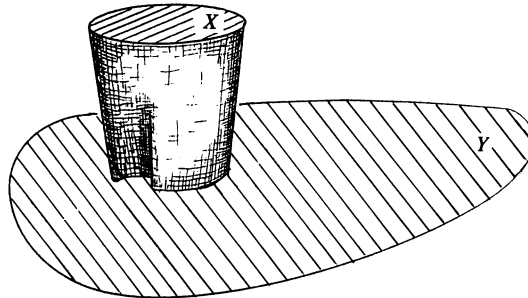


Figure I-2. Mapping cylinder.

It is often of interest to know when a function $M_f \rightarrow Z$, taking a mapping cylinder into another space Z , is continuous. This is usually quite easy to check by use of the following simple fact. The proof is an easy application of the definition of the quotient topology on a mapping cylinder, and is left to the reader.

13.18. Proposition. *A function $M_f \rightarrow Z$ is continuous $\Leftrightarrow$ the induced functions $X \times I \rightarrow Z$ and $Y \rightarrow Z$ are both continuous.* $\square$

Here is a result we shall need later.

13.19. Proposition. *If $f: X \rightarrow Y$ is an identification map and K is a locally compact Hausdorff space then $f \times 1: X \times K \rightarrow Y \times K$ is an identification map.*

PROOF. Suppose that $g: Y \times K \rightarrow W$ and let $h = g \circ (f \times 1): X \times K \rightarrow W$. Then, by Proposition 13.5, it suffices to prove that h continuous $\Rightarrow g$ continuous. Let $U \subset W$ be open and suppose that $g(y_0, k_0) \in U$. Let $f(x_0) = y_0$. Then $h(x_0, k_0) = g(y_0, k_0) \in U$. Therefore there is a compact neighborhood N of k_0 such that $h(x_0 \times N) \subset U$. Put $A = \{y \in Y \mid g(y \times N) \subset U\}$. Then $y_0 \in A$ and it suffices to show that A is open. Thus it suffices to show that $f^{-1}(A)$ is open. Now

$$f^{-1}(A) = \{x \in X \mid h(x \times N) = g(f(x) \times N) \subset U\}$$

and so $X - f^{-1}(A) = \pi_X(h^{-1}(W - U) \cap (X \times N))$ is closed by Proposition 8.2. $\square$

PROBLEMS

1. $\diamond$ If $f: X \rightarrow A$ and $g: Y \rightarrow B$ are open identification maps, show that $f \times g: X \times Y \rightarrow A \times B$ is also an open identification map.
2. If X, Y are normal, $A \subset X$ is closed, and $f: A \rightarrow Y$ is a map, show that $Y \cup_f X$ is normal.
3. If $f: X \rightarrow Y$ is a map between Hausdorff spaces, show that M_f and C_f are Hausdorff.
4. There are four common definitions of the torus $\mathbf{T}^2$:
 - (1) as $\mathbf{R}^2/\mathbf{Z}^2$, i.e., the plane modulo the equivalence relation $(x, y) \sim (u, w) \Leftrightarrow x - u$ and $y - w$ are both integers;
 - (2) as a square with opposite edges identified (see Figure I-3);
 - (3) as the product $\mathbf{S}^1 \times \mathbf{S}^1$; and
 - (4) as the “anchor ring,” the surface of revolution obtained by rotating a circle about an axis in its plane and disjoint from it.
 Show that these are all homeomorphic to one another.
5. The “Klein bottle” $\mathbf{K}^2$ is a square with opposite vertical edges identified in the same direction and opposite horizontal edges identified in the opposite direction (see Figure I-3). Consider the space (denoted by $\mathbf{P}^2 \# \mathbf{P}^2$) resulting from an annulus by identifying antipodal points on the outer circle, and also identifying antipodal points on the inner circle. Show that $\mathbf{K}^2 \approx \mathbf{P}^2 \# \mathbf{P}^2$.

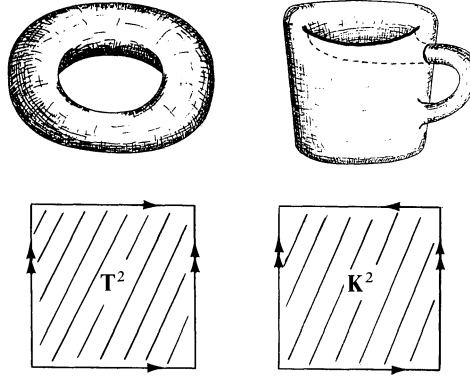


Figure I-3. The torus (left) and Klein bottle (right).

6. Consider the real line $\mathbf{R}$, with the equivalence relation $x \sim y \Leftrightarrow x - y$ is rational. Show that $\mathbf{R}/\sim$ has an uncountable number of points, but its topology is the trivial one.
7. Consider the real line $\mathbf{R}$ and the integers $\mathbf{Z}$. Let $A = \mathbf{R}/\mathbf{Z}$ (the identification of the subspace $\mathbf{Z}$ to a point). Also consider the subspace B of the plane which is the union of the circles of radius $1/n$ ($n = 1, 2, \dots$) in the upper half plane all tangent to the real line at the origin. Also consider the subspace C of the plane which is the union of the circles of radius n ($n = 1, 2, \dots$) in the upper half plane all tangent to the real line at the origin. Finally, consider the space $D = \mathbf{S}^1/N$ where $\mathbf{S}^1$ is the unit circle in the complex numbers and $N = \{e^{in/n} | n = 1, 2, \dots\} \cup \{1\}$. Which of these four spaces A, B, C, D are homeomorphic to which others of them?
8. Let (X, x_0) and (Y, y_0) be “pointed spaces,” i.e., spaces with distinguished “base” points. Define the “one-point union” $X \vee Y$ to be the quotient space of the topological sum $X + Y$ by the equivalence relation identifying x_0 with y_0 . Show that $X \vee Y \approx X \times \{y_0\} \cup \{x_0\} \times Y$, where the latter is regarded as a subspace of $X \times Y$.

14. Homotopy

A homotopy is a family of mappings parametrized by the unit interval. This notion is of central importance in topology. Here we lay down the basic definitions and properties of this concept.

14.1. Definition. If X and Y are spaces then a *homotopy* of maps from X to Y is a map $F: X \times I \rightarrow Y$, where $I = [0, 1]$.

Two maps $f_0, f_1: X \rightarrow Y$ are said to be *homotopic* if there exists a homotopy $F: X \times I \rightarrow Y$ such that $F(x, 0) = f_0(x)$ and $F(x, 1) = f_1(x)$ for all $x \in X$.

The relation “ f is homotopic to g ” is an equivalence relation on the set of all maps from X to Y (see Definition 14.11) and is denoted by $f \simeq g$. The following is elementary:

14.2. Proposition. *If $f, g: X \rightarrow Y, h: X' \rightarrow X$ and $k: Y \rightarrow Y'$ then*

$$f \simeq g \Rightarrow f \circ h \simeq g \circ h \quad \text{and} \quad k \circ f \simeq k \circ g. \quad \square$$

14.3. Definition. A map $f: X \rightarrow Y$ is said to be a *homotopy equivalence* with *homotopy inverse* g if there is a map $g: Y \rightarrow X$ such that $g \circ f \simeq 1_X$ and $f \circ g \simeq 1_Y$. This relationship is denoted by $X \simeq Y$. One also says, in this case, that X and Y have the *same homotopy type*.

This is an equivalence relation between spaces, since, if $h: Y \rightarrow Z$ is another homotopy equivalence with homotopy inverse k then

$$(gk)(hf) = g(kh)f \simeq g1_Y f = gf \simeq 1_X$$

and similarly for the opposite composition.

14.4. Definition. A space is said to be *contractible* if it is homotopy equivalent to the one-point space.

14.5. Proposition. *A space X is contractible $\Leftrightarrow$ the identity map $1_X: X \rightarrow X$ is homotopic to a map $r: X \rightarrow X$ whose image is a single point.*

PROOF. Let $Y = \{x_0\} = \text{im}(r)$. Then we have the inclusion map $i: Y \rightarrow X$ and the retraction $r: X \rightarrow Y$. Now $r \circ i = 1_Y$ and $i \circ r \simeq 1_X$ by assumption. The converse is also easy. $\square$

14.6. Example. Consider euclidean space $X = \mathbf{R}^n$ and the homotopy $F: X \times I \rightarrow X$ given by $F(x, t) = tx$. This is a homotopy between $f_1 = 1_X$ and f_0 , which is the map taking everything to $\{0\}$. Consequently, $\mathbf{R}^n$ is contractible. Note that each f_t is onto for $t > 0$ but that, suddenly, f_0 is far from onto. This may challenge the intuition of some readers.

14.7. Example. Consider the unit sphere S^{n-1} in $\mathbf{R}^n$ and the punctured euclidean space $\mathbf{R}^n - \{0\}$. Let $i: S^{n-1} \rightarrow \mathbf{R}^n - \{0\}$ be the inclusion and $r: \mathbf{R}^n - \{0\} \rightarrow S^{n-1}$ be the central projection $r(x) = x/\|x\|$. Then $r \circ i = 1$ and $i \circ r \simeq 1$ where the latter homotopy is given by $F: (\mathbf{R}^n - \{0\}) \times I \rightarrow \mathbf{R}^n - \{0\}$, where $F(x, t) = tx + (1-t)x/\|x\|$. Thus $S^{n-1} \simeq \mathbf{R}^n - \{0\}$.

These two examples illustrate and suggest the following:

14.8. Definition. A subspace A of X is called a *strong deformation retract* of

X if there is a homotopy $F: X \times I \rightarrow X$ (called a deformation) such that:

$$\begin{aligned} F(x, 0) &= x, \\ F(x, 1) &\in A, \\ F(a, t) &= a \quad \text{for } a \in A \text{ and all } t \in I. \end{aligned}$$

It is just a *deformation retract* if the last equation is required only for $t = 1$.

As in the examples, a deformation retract A of a space X is homotopically equivalent to X .

Is the sphere S^n contractible? Our intuition tells us the answer is “no” but, in fact, this is quite difficult to prove. This is one type of question which algebraic topology is equipped to answer, and we will answer it, and many more such questions, in later pages.

14.9. Example. If $f: X \rightarrow Y$ is a map then the canonical map $r: M_f \rightarrow Y$ is a strong deformation retraction, as the reader can verify (see the end of the proof of Theorem 14.18). Hence $M_f \simeq Y$. Thus, the mapping cylinder allows replacing “up to homotopy” the arbitrary map f by the inclusion $X \hookrightarrow M_f$.

14.10. Definition. If $A \subset X$ then a homotopy $F: X \times I \rightarrow Y$ is said to be *relative to A* (or *rel A*) if $F(a, t)$ is independent of t for $a \in A$. A homotopy that is *rel X* is said to be a *constant* homotopy.

Two homotopies of X into Y can be “concatenated” if the first ends where the second begins, by going through the first at twice the normal speed and then the second at that speed. We will now study this construction. The reader should note the important special case in which X is a single point and so the homotopies are simply paths in Y . It might help the reader’s understanding if he draws pictures, in this case, for all the basic homotopies produced in Propositions 14.13, 14.15, and 14.16.

14.11. Definition. If $F: X \times I \rightarrow Y$ and $G: X \times I \rightarrow Y$ are two homotopies such that $F(x, 1) = G(x, 0)$ for all x , then define a homotopy $F * G: X \times I \rightarrow Y$, which is called *the concatenation of F and G* , by

$$(F * G)(x, t) = \begin{cases} F(x, 2t) & \text{if } t \leq \frac{1}{2}, \\ G(x, 2t - 1) & \text{if } t \geq \frac{1}{2}. \end{cases}$$

(See Figure I-4.)

One does not have to combine these homotopies at $t = \frac{1}{2}$. We can do it at any point and with arbitrary speed:

14.12. Lemma (Reparametrization Lemma). *Let ϕ_1 and ϕ_2 be maps $(I, \partial I) \rightarrow (I, \partial I)$ which are equal on ∂I . (Note the case where one of these is the*

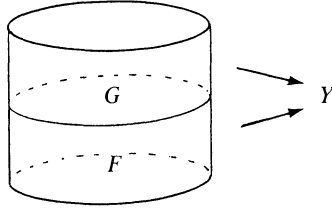


Figure I-4. Concatenation of homotopies.

identity.) Let $F: X \times I \rightarrow Y$ be a homotopy and let $G_i(x, t) = F(x, \phi_i(t))$ for $i = 1, 2$. Then $G_1 \simeq G_2 \text{ rel } X \times \partial I$.

PROOF. Define $H: X \times I \times I \rightarrow Y$ by $H(x, t, s) = F(x, s\phi_2(t) + (1-s)\phi_1(t))$. Then

$$\begin{aligned} H(x, t, 0) &= F(x, \phi_1(t)) = G_1(x, t), \\ H(x, t, 1) &= F(x, \phi_2(t)) = G_2(x, t), \\ H(x, 0, s) &= F(x, \phi_1(0)) = G_1(x, 0), \\ H(x, 1, s) &= F(x, \phi_2(1)) = G_2(x, 1), \end{aligned}$$

with the last two equations coming from $\phi_1(0) = \phi_2(0)$ and $\phi_1(1) = \phi_2(1)$. $\square$

We shall use C to denote a constant homotopy, whichever one makes sense in the current context. For example $F * C$ is concatenation with the constant homotopy C for which $C(x, t) = F(x, 1)$, but use of $C * F$ will imply the one for which $C(x, t) = F(x, 0)$.

14.13. Proposition. We have $F * C \simeq F \text{ rel } X \times \partial I$, and, similarly, $C * F \simeq F \text{ rel } X \times \partial I$.

PROOF. This follows from Lemma 14.12 by letting $\phi_1(t) = 2t$ for $t \leq \frac{1}{2}$ and $= 1$ for $t \geq \frac{1}{2}$, and $\phi_2(t) = t$ in the first case, and $\phi_1(t) = 0$ for $t \leq \frac{1}{2}$ and $= 2t - 1$ for $t \geq \frac{1}{2}$ and $\phi_2(t) = t$ in the second case. $\square$

We define the “inverse” F^{-1} of a homotopy F to be this homotopy with t running backward. Note that this has nothing to do with the inverse of the map (which probably does not exist anyway).

14.14. Definition. If $F: X \times I \rightarrow Y$ is a homotopy, then we define $F^{-1}: X \times I \rightarrow Y$ by $F^{-1}(x, t) = F(x, 1 - t)$.

14.15. Proposition. For a homotopy F we have $F * F^{-1} \simeq C \text{ rel } X \times \partial I$ where $C(x, t) = F(x, 0)$ for all x and t ; i.e., C is a constant homotopy.

PROOF. This follows from Lemma 14.12 by letting $\phi_1(t) = 2t$ for $t \leq \frac{1}{2}$, $\phi_1(t) = 2 - 2t$ for $t \geq \frac{1}{2}$, and $\phi_2(t) = 0$ for all t . $\square$

14.16. Proposition. *For any homotopies F, G, H for which the concatenations $F * G$ and $G * H$ are defined, we have $(F * G) * H \simeq F * (G * H) \text{ rel } X \times \partial I$.*

PROOF. Again, this is an easy application of the Reparametrization Lemma (Lemma 14.12). $\square$

14.17. Proposition. *For homotopies F_1, F_2, G_1 , and G_2 , if $F_1 \simeq F_2 \text{ rel } X \times \partial I$ and $G_1 \simeq G_2 \text{ rel } X \times \partial I$ then $F_1 * G_1 \simeq F_2 * G_2 \text{ rel } X \times \partial I$.*

PROOF. If $H: X \times I \times I \rightarrow Y$ and $K: X \times I \times I \rightarrow Y$ are the homotopies giving $F_1 \simeq F_2$ and $G_1 \simeq G_2$, respectively, then it is easy to check that $H * K$ is the required homotopy of homotopies. (The reader should fill in the details here.) $\square$

Note that all of the discussion of concatenation of homotopies goes through with no difficulties for the cases in which all homotopies are relative to some subspace $A \subset X$ or are homotopies of pairs $(X, A) \rightarrow (Y, B)$.

It follows from the stated results that homotopy between maps of pairs $(X, A) \rightarrow (Y, B)$ is an equivalence relation. The set of homotopy classes of these maps is commonly denoted by $[X, A; Y, B]$, or just $[X; Y]$ if $A = \emptyset$.

We will now prove that the homotopy type of a mapping cylinder or cone depends only on the homotopy class of the map.

14.18. Theorem. *If $f_0 \simeq f_1: X \rightarrow Y$ then $M_{f_0} \simeq M_{f_1} \text{ rel } X + Y$ and $C_{f_0} \simeq C_{f_1} \text{ rel } Y + \text{vertex}$.*

PROOF. The part for the cone follows from that for the mapping cylinder. Let $F: X \times I \rightarrow Y$ be the given homotopy between f_0 and f_1 . Define $h: M_{f_0} \rightarrow M_{f_1}$ by $h(y) = y$ for $y \in Y$ and

$$h(x, t) = \begin{cases} F(x, 2t) & \text{for } t \leq \frac{1}{2}, \\ (x, 2t - 1) & \text{for } \frac{1}{2} \leq t. \end{cases}$$

Note that $h(x, \frac{1}{2}) = F(x, 1) = f_1(x) = (x, 0)$. To prove continuity, we only have to show that the compositions $Y \rightarrow M_{f_1}$ and $X \times I \rightarrow M_{f_1}$ are continuous by Proposition 13.18, but this is trivial.

Define $k: M_{f_1} \rightarrow M_{f_0}$ in the analogous fashion. Then the composition $kh: M_{f_0} \rightarrow M_{f_0}$ is the identity on Y and, on the cylinder portion, it is $F * (F^{-1} * E)$ where $E: X \times I \rightarrow M_{f_0}$ is induced by the identity on $X \times I \rightarrow X \times I$; see Figure I-5. This is homotopic to the identity $\text{rel } X \times \{1\} + Y$. Similarly for hk . It remains to check the continuity of this homotopy. We have described a homotopy $M_{f_0} \times I \rightarrow M_{f_0}$. For continuity, it is sufficient to know that $M_{f_0} \times I \approx M_{f_0 \times I}$ because then we only have to check continuity of the

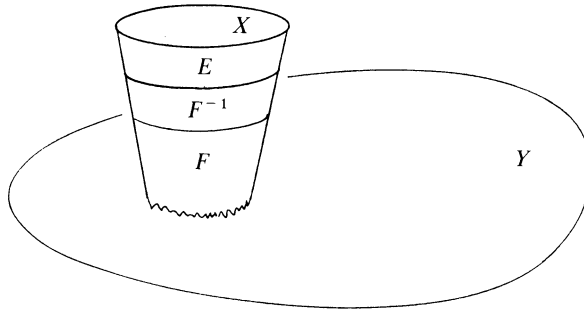


Figure I-5. Deformation of a mapping cylinder.

composition $(X \times I + Y) \times I \rightarrow M_{f_0} \times I \rightarrow M_{f_0}$, and that is trivial. (On $Y \times I$ it is the constant homotopy and on $X \times I \times I$ it results from $F \ast (F^{-1} \ast E) \simeq E \text{ rel } X \times \partial I$.) That is, it suffices to show that $M_{f_0} \times I$ has the identification topology from the map $f_0 \times I$. But that is a consequence of Proposition 13.19. $\square$

We conclude this section by studying the effect on mapping cones of changing the target space by a homotopy equivalence.

Let $f: X \rightarrow Y$. If $\phi: Y \rightarrow Y'$ is a map then there is the induced map $F: M_f \rightarrow M_{\phi \circ f}$ induced from ϕ on Y and the identity on $X \times I$.

14.19. Theorem. *If $\phi: Y \rightarrow Y'$ is a homotopy equivalence then so is $F: (M_f, X) \rightarrow (M_{\phi \circ f}, X)$ and hence so is $F: C_f \rightarrow C_{\phi \circ f}$.*

PROOF. Let $\psi: Y' \rightarrow Y$ be a homotopy inverse of ϕ and let $G: M_{\phi \circ f} \rightarrow M_{\psi \circ \phi \circ f}$ be the map induced by ψ on Y' and the identity on $X \times I$. The composition $GF: M_f \rightarrow M_{\psi \circ \phi \circ f}$ is induced from $\psi \circ \phi: Y \rightarrow Y$ and the identity on $X \times I$. Let $H: Y \times I \rightarrow Y$ be a homotopy from 1 to $\psi \circ \phi$; i.e., $H(y, 0) = y$ and $H(y, 1) = \psi \phi(y)$. By the proof of Theorem 14.18 there is the homotopy equivalence $h: M_f \rightarrow M_{\psi \circ \phi \circ f} \text{ rel } X$ given by $h(y) = y$ and

$$h(x, t) = \begin{cases} H(f(x), 2t) & \text{for } t \leq \frac{1}{2}, \\ (x, 2t - 1) & \text{for } t \geq \frac{1}{2}. \end{cases}$$

We claim that $h \simeq GF \text{ rel } X$. Indeed, the homotopy H can be extended to $M_f \times I \rightarrow M_{\psi \circ \phi \circ f}$ by putting

$$H((x, s), t) = \begin{cases} H(f(x), 2s + t) & \text{for } 2s + t \leq 1, \\ (x, (2s + t - 1)/(t + 1)) & \text{for } 2s + t \geq 1. \end{cases}$$

Then $H((x, s), 0) = h(x, s)$, $H(y, 0) = y$, and $H((x, s), 1) = (x, s)$, $H(y, 1) = \psi \phi(y)$, so that $H(\cdot, 0) = h$ and $H(\cdot, 1) = GF$.

We conclude that GF is a homotopy equivalence, since h is one. Likewise, $F'G$ is a homotopy equivalence, where $F': M_{\psi \circ \phi \circ f} \rightarrow M_{\phi \circ \psi \circ \phi \circ f}$ is defined similarly to F .

If k is a homotopy inverse of GF then $GFk \simeq 1$. If k' is a homotopy inverse of $F'G$ then $k'F'G \simeq 1$. Thus G has a homotopy right inverse $R = Fk$ and also a homotopy left inverse $L = k'F'$. That is, $LG \simeq 1 \simeq GR$. Then $R = 1 \circ R \simeq (LG)R = L(GR) \simeq L \circ 1 = L$, and so $R \simeq L$ is a homotopy inverse of G . Therefore, G is a homotopy equivalence. Since G and GF are homotopy equivalences, so is F . (Explicitly, if l is a homotopy inverse of G then $FkG \simeq (lG)FkG = l(GFk)G \simeq lG \simeq 1$ so that kG is a homotopy inverse of F , since $kGF \simeq 1$ by the definition of k .) $\square$

PROBLEMS

1. Let $S^2 \vee S^1$ be the “one-point union” of a 2-sphere and a circle; see Problem 8 of Section 13. Let $S^2 \cup A$ denote the union of the unit 2-sphere and the line segment joining the north and south poles. Show that these spaces are homotopically equivalent.
2. Show that the union of a 2-sphere and a flat unit 2-cell through the origin is homotopically equivalent to the one-point union of two 2-spheres.
3. Show that the union of a standard 2-torus with two disks, one spanning a latitudinal circle and the other spanning a longitudinal circle of the torus, is homotopically equivalent to a 2-sphere.
4. Show that the projective plane is homeomorphic to the mapping cone of the map $z \mapsto z^2$ of the unit circle in the complex numbers to itself.
5. Consider the mapping cone of the map f of the unit circle in the complex numbers to itself, given by $f(z) = z^4$ for z in the upper semicircle and by $f(z) = \bar{z}^2$ for z in the lower semicircle. Show that this space is contractible.
6. The “dunce cap” space is the quotient of a triangle (and interior) obtained by identifying all three edges in an inconsistent manner. That is, if the vertices of the triangle are p, q, r then we identify the line segment (p, q) with (q, r) and with (p, r) in the orientation indicated by the order given of the vertices. (See Figure I-6.) Show that the dunce cap is contractible. (*Hint*: Describe this space as the mapping cone of a certain map from S^1 to itself, and study this map.)

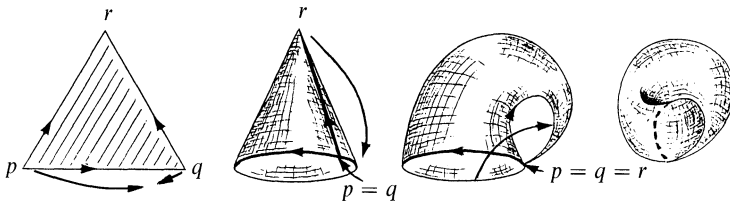


Figure I-6. The dunce cap.

7. If $\phi: X' \rightarrow X$ and $f: X \rightarrow Y$ are maps, define an induced map $F: C_{f \circ \phi} \rightarrow C_f$. If ϕ is a homotopy equivalence then show that F is a homotopy equivalence.
8. Show that a retract of a contractible space is contractible.
9. For any two maps $f, g: X \rightarrow S^n$ such that $f(x) \neq -g(x)$ for all x , show that $f \simeq g$.

15. Topological Groups

Topological groups, spaces which are also groups in the algebraic sense, form a rich territory for important examples in topology and geometry. Here we shall develop the roots of the theory behind them.

15.1. Definition. A *topological group* is a Hausdorff topological space G together with a group structure on G such that:

- (1) group multiplication $(g, h) \mapsto gh$ of $G \times G \rightarrow G$ is continuous; and
- (2) group inversion $g \mapsto g^{-1}$ of $G \rightarrow G$ is continuous.

15.2. Definition. A *subgroup* H of a topological group G is a subspace which is also a subgroup in the algebraic sense.

15.3. Definition. If G and G' are topological groups then a *homomorphism* $f: G \rightarrow G'$ is a group homomorphism which is also continuous.

15.4. Definition. If G is a topological group and $g \in G$ then *left translation* by g is the map $L_g: G \rightarrow G$ given by $L_g(h) = gh$. Similarly *right translation* by g is the map $R_g: G \rightarrow G$ given by $R_g(h) = hg^{-1}$.

15.5. Proposition. In a topological group G we have $L_g \circ L_h = L_{gh}$ and $R_g \circ R_h = R_{gh}$. Moreover, both L_g and R_g are homeomorphisms as is conjugation by g ($h \mapsto ghg^{-1}$) and inversion ($h \mapsto h^{-1}$).

PROOF. The first statement is a trivial computation, and implies that $L_{g^{-1}} = L_g^{-1}$ and similarly for right translation, and it follows that these are homeomorphisms. Conjugation $h \mapsto ghg^{-1}$ is the same as $R_g \circ L_g$ and so is a homeomorphism. Inversion is continuous by assumption and is its own inverse, and thus is a homeomorphism. $\square$

In a topological group G , if A, B are subsets then we let $AB = \{ab | a \in A, b \in B\}$ and $A^{-1} = \{a^{-1} | a \in A\}$.

15.6. Definition. A subset A of a topological group is called *symmetric* if $A = A^{-1}$.

15.7. Proposition. In a topological group G with unity element e , the symmetric neighborhoods of e form a neighborhood basis at e .

PROOF. If U is any neighborhood of e then so is U^{-1} , and hence so is $U \cap U^{-1}$, which is symmetric. $\square$

It is an easy exercise using the continuity of multiplication to see the following two results:

15.8. Proposition. *If G is a topological group and $g \in G$, and U is any neighborhood of g , then there is a symmetric neighborhood V of e such that $VgV^{-1} \subset U$.* $\square$

15.9. Proposition. *If G is a topological group and U is any neighborhood of e and n is any positive integer, then there exists a symmetric neighborhood V of e such that $V^n \subset U$.* $\square$

15.10. Proposition. *If H is any subgroup of a topological group G then $\bar{H}$ is also a subgroup of G . If H is a normal subgroup then so is $\bar{H}$.*

PROOF. It follows from continuity of inversion and multiplication that $\bar{H}^{-1} \subset \bar{H}$ and $\bar{H}\bar{H} \subset \bar{H}$ so that $\bar{H}$ is a subgroup. If H is a normal subgroup and $g \in G$ then continuity also implies that $g\bar{H}g^{-1} \subset \bar{H}$ and the opposite inclusion follows by applying this formula to g^{-1} . $\square$

15.11. Proposition. *If G is a topological group and H is a closed subgroup then the space G/H of left cosets of H in G , with the topology induced by the canonical map $\pi: G \rightarrow G/H$, is a Hausdorff space. Moreover, π is open and continuous.*

PROOF. If $U \subset G$ is open then $\pi^{-1}\pi(U) = UH = \bigcup \{Uh | h \in H\}$ is a union of open sets and so is open. By definition of the quotient topology it follows that $\pi(U)$ is open, proving the last statement. To see that G/H is Hausdorff, suppose that $g_1H \neq g_2H$ (hence representing different points in G/H). This is the same as saying that $g_1^{-1}g_2 \notin H$. Since $G-H$ is an open set containing $g_1^{-1}g_2$, Proposition 15.8 implies that there is a symmetric open neighborhood U of e such that $(Ug_1^{-1}g_2U) \cap H = \emptyset$. Thus $g_1^{-1}g_2U \cap UH = \emptyset$ which is the same as $g_2U \cap g_1UH = \emptyset$ which implies in turn that $g_2UH \cap g_1UH = \emptyset$. This shows that $\pi(g_i) \in g_iUH$ which are disjoint open sets in G/H . $\square$

15.12. Proposition. *If H is a closed normal subgroup of the topological group G then G/H , with the quotient topology, is a topological group.*

PROOF. G/H is Hausdorff by Proposition 15.11 and it remains to show that the group operations are continuous. Consider the following diagram, where the horizontal arrows are group multiplications:

$$\begin{array}{ccc} G \times G & \longrightarrow & G \\ \downarrow \pi \times \pi & & \downarrow \pi \\ G/H \times G/H & \longrightarrow & G/H. \end{array}$$

An easy consequence of the fact (Proposition 15.11) that π is open is that $\pi \times \pi$ is an identification map (see Section 13, Problem 1). Taking an open set in G/H (the lower right), we must show its inverse image on the lower left is open. But that is the same as showing that its inverse image in $G \times G$ is open. But this inverse image is the same as that via the top and right maps, and that is open since those maps are continuous. A similar argument gives the continuity of inversion in G/H . $\square$

The most important class of topological groups is that of the so-called Lie groups which also carry a differentiable structure. We will discuss differentiable structures in Chapter II. Here, we will only discuss some of the important “classical Lie groups.”

The set M_n of $n \times n$ -matrices is just a euclidean space of dimension n^2 . The determinant function $M_n \rightarrow \mathbf{R}$ is continuous since it is just a polynomial in the matrix coefficients. Thus the inverse image of $\{0\}$ is a closed set. Its complement is the set of nonsingular matrices, and this forms a group under multiplication. It is called the “general linear group” and is denoted by $\mathbf{GL}(n, \mathbf{R})$. It is an open subset of euclidean n^2 -space and that is the topology it is given. Matrix multiplication is given by polynomials in the coefficients and so is continuous. Matrix inversion is a rational function of the coefficients by Cramer’s rule and so that is continuous. Thus $\mathbf{GL}(n, \mathbf{R})$ is a topological group.

In the same way, the general linear group $\mathbf{GL}(n, \mathbf{C})$ over the complex numbers can be seen to be a topological group.

The special linear group $\mathbf{SL}(n, \mathbf{R})$ is the subgroup of $\mathbf{GL}(n, \mathbf{R})$ consisting of matrices of determinant 1, and similarly for $\mathbf{SL}(n, \mathbf{C})$ over the complexes.

Similarly, the general linear group $\mathbf{GL}(n, \mathbf{H})$ over the quaternions is a topological group, although, in this case, the argument is a little harder since quaternionic matrices lack a determinant function. (See Problem 12.)

The set $\mathbf{O}(n)$ of orthogonal (real) matrices forms a subgroup of $\mathbf{GL}(n, \mathbf{R})$ and it is a closed subset, since it is defined via continuous relations ($AA^t = I$). Since the coefficients of an orthogonal matrix are bounded by 1 in absolute value, $\mathbf{O}(n)$ is a bounded closed subset of euclidean n^2 -space, and hence is compact by Corollary 8.7.

Similarly, the set $\mathbf{U}(n)$ of unitary matrices ($AA^* = I$) is a compact subgroup of $\mathbf{GL}(n, \mathbf{C})$.

The quaternionic analogue of the orthogonal and unitary groups is called the symplectic group $\mathbf{Sp}(n)$. Its elements are quaternionic matrices A such that $AA^* = I$, where A^* is the quaternionic conjugate transpose of A , conjugation meaning reversal of all three imaginary components. This group is a compact subgroup of $\mathbf{GL}(n, \mathbf{H})$.

These three classes of examples are called the “classical Lie groups.”

Note that the map $\mathbf{GL}(n, \mathbf{R}) \times \mathbf{R}^n \rightarrow \mathbf{R}^n$ is given by polynomials in the coefficients of the matrix and the vector, and so is continuous.

An orthogonal matrix $A \in \mathbf{O}(n)$, as a transformation of euclidean n -space, preserves lengths of vectors, and so it is a map of the sphere $\mathbf{S}^{n-1}$ to itself. We can regard $\mathbf{O}(n-1)$ as the subgroup of $\mathbf{O}(n)$ fixing the last coordinate.

Consider the point $(0, 0, \dots, 0, 1)$. This point is left fixed by $\mathbf{O}(n-1)$. We can map $\mathbf{O}(n)$ into $\mathbf{S}^{n-1}$ by taking a matrix into where it moves the point $(0, 0, \dots, 0, 1)$. That is, we define the map

$$f: \mathbf{O}(n) \rightarrow \mathbf{S}^{n-1} \quad \text{by} \quad f(A) = A(0, 0, \dots, 0, 1)^t.$$

If $B \in \mathbf{O}(n-1)$ then clearly $f(AB) = f(A)$. This means that the map f factors through the left coset space $\mathbf{O}(n)/\mathbf{O}(n-1)$. A short computation will show that the induced map $\mathbf{O}(n)/\mathbf{O}(n-1) \rightarrow \mathbf{S}^{n-1}$ is one-one onto and continuous. Since this is a one-one mapping of a compact space onto a Hausdorff space it is a homeomorphism by Theorem 7.8. Let us abstract these observations.

15.13. Definition. If G is a topological group and X is space, then an *action* of G on X is a map $G \times X \rightarrow X$, with the image of (g, x) being denoted by $g(x)$, such that:

- (1) $(gh)(x) = g(h(x))$; and
- (2) $e(x) = x$.

For a point $x \in X$, the set $G(x) = \{g(x) | g \in G\}$ is called the *orbit* of x , and the subgroup $G_x = \{g \in G | g(x) = x\}$ is called the *isotropy* or *stability* group at x . The action is said to be *transitive* if there is only one orbit, the whole space X . The action is said to be *effective* if $(g(x) = x \text{ for all } x) \Rightarrow g = e$, the identity element of G .

Note that, in describing $G \times X \rightarrow X$ as a “map,” we are assuming it to be continuous. The following, then, is the general setting in which our comments on $\mathbf{O}(n)$ acting on $\mathbf{S}^{n-1}$ lie.

15.14. Proposition. *If G is a compact topological group acting on the Hausdorff space X and G_x is the isotropy group at x , then the map $\phi: G/G_x \rightarrow G(x)$ given by $gG_x \mapsto g(x)$ is a homeomorphism.*

PROOF. If $g_1(x) = g_2(x)$ then $g_1^{-1}g_2 \in G_x$ and so $g_1G_x = g_2G_x$, showing that ϕ is one-one onto $G(x)$. It is continuous by the definition of the quotient topology on G/G_x , and the result then follows from Theorem 7.8. $\square$

Just as with the case of $\mathbf{O}(n)$, $\mathbf{U}(n)$ acts on $\mathbf{S}^{2n-1}$, and it is transitive because one can find a unitary matrix moving any vector of length 1 into any other. The isotropy group at $(0, 0, \dots, 0, 1)$ is $\mathbf{U}(n-1)$, and so $\mathbf{U}(n)/\mathbf{U}(n-1) \approx \mathbf{S}^{2n-1}$.

Similarly $\mathbf{Sp}(n)/\mathbf{Sp}(n-1) \approx \mathbf{S}^{4n-1}$.

More generally, if we let (as is usual) $V_{n,k}$ denote the “Stiefel manifold” of k -frames in n -space (a k -frame being an orthonormal set of k vectors in n -space), then $\mathbf{O}(n)$ acts transitively on $V_{n,k}$ with an isotropy group $\mathbf{O}(n-k)$, and so $\mathbf{O}(n)/\mathbf{O}(n-k) \approx V_{n,k}$. The reader can make analogous observations for the unitary and symplectic cases.

Some other matrix groups are obtained by restriction to matrices of determinant 1:

$\mathbf{SO}(n) = \{A \in \mathbf{O}(n) | \det(A) = 1\}$ = the special orthogonal group,

$\mathbf{SU}(n) = \{A \in \mathbf{U}(n) | \det(A) = 1\}$ = the special unitary group.

There is no analogue in the symplectic case.

With appropriate restrictions, these groups also act transitively on spheres and we get

$$\mathbf{SO}(n)/\mathbf{SO}(n-1) \approx \mathbf{S}^{n-1} \quad \text{for } n \geq 2,$$

$$\mathbf{SU}(n)/\mathbf{SU}(n-1) \approx \mathbf{S}^{2n-1} \quad \text{for } n \geq 2.$$

Similar results can be obtained for the Stiefel manifolds, as the reader can verify.

PROBLEMS

1. If G is a topological group and G_0 is the component containing the identity element, then show that G_0 is a closed normal subgroup of G .
2. If $\phi: G \rightarrow H$ is an onto homomorphism of topological groups, show that the kernel K of ϕ is a closed normal subgroup of G . If, moreover, G is compact, show that $G/K \approx H$ as topological groups.
3. If $g \in G$, a compact topological group, and $A = \{g^n | n = 0, 1, 2, \dots\}$, then show that $\bar{A}$ is a subgroup of G . Is this true without compactness of G ?
4. If G is a compact topological group, then show that every neighborhood of e contains a neighborhood V which is invariant under conjugation (i.e., $gVg^{-1} = V$ for all $g \in G$).
5. $\diamond$ If G is a topological group and H is a closed subgroup, show that if H and G/H are both connected then so is G .
6. If G is a topological group acting on the space X and if we put

$$H = \{h \in G | \forall x \in X, h(x) = x\},$$

then show that H is a closed normal subgroup of G .

7. $\diamond$ Show that $\mathbf{SO}(2) \approx \mathbf{S}^1$, $\mathbf{SU}(2) \approx \mathbf{S}^3$, and $\mathbf{Sp}(1) \approx \mathbf{S}^3$ (as spaces).
8. $\diamond$ Show that $\mathbf{SO}(n)$ is connected. (*Hint*: Use Problem 5.) Further, show that $\mathbf{SO}(n)$ is the component of $\mathbf{O}(n)$ containing the identity.
9. $\diamond$ Show that $\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are both connected and that $\mathbf{U}(n)/\mathbf{SU}(n) \approx \mathbf{S}^1$.
10. Show that the center $\{h \in G | \forall g \in G, hg = gh\}$ of a topological group G is closed.
11. Show that real projective n -space $\mathbf{RP}^n \approx \mathbf{O}(n+1)/(\mathbf{O}(n) \times \mathbf{O}(1))$ and that complex projective n -space $\mathbf{CP}^n \approx \mathbf{U}(n+1)/(\mathbf{U}(n) \times \mathbf{U}(1))$.
12. $\diamond$ Show that $\mathbf{GL}(n, \mathbf{H})$ is open in $M_n(\mathbf{H})$.
13. Consider the multiplicative group of all upper diagonal 2×2 matrices of deter-

- minant 1. Show that the conclusion of Problem 4 is false for this (noncompact) topological group.
14. Show that a topological group is regular. (*Hint*: Let U, V be symmetric open neighborhoods of e such that $V^2 \subset U$ and deduce that $\bar{V} \subset U$.)
15. Prove Propositions 15.8 and 15.9.

16. Convex Bodies

In topology, we often need to know that certain familiar objects are homeomorphic. For example, we shall have reason to want to know that a disk in euclidean space is homeomorphic to a cube, and to a cylinder, and a simplex (the analogue of a tetrahedron), and so on. In this section we give a general result that provides a unified proof of these special cases and many others.

16.1. Definition. A *convex body* in $\mathbf{R}^n$ is a closed set $C \subset \mathbf{R}^n$ with the property that whenever $p, q \in C$ the line segment between p and q is contained in C . The *boundary* of C is $\partial C = C - \text{int}(C)$.

16.2. Proposition. If $C \subset \mathbf{R}^n$ is a convex body and $0 \in \text{int}(C)$ then any ray from the origin intersects ∂C in at most one point.

PROOF. Suppose R is a ray from the origin and $p, q \in R \cap C$, with neither p nor q being the origin. Suppose q is further from the origin than p . Since the origin is assumed to lie in $\text{int}(C)$ there is a ball B about the origin completely contained in C . Then consider the union of all line segments from points in B to q (the cone on B subtended from q). The point p is clearly in the interior of this cone, and the cone is contained completely in C , since C is convex, and so p must be in $\text{int}(C)$. $\square$

16.3. Proposition. Let $C \subset \mathbf{R}^n$ be a compact convex body with $0 \in \text{int}(C)$. Then the function $f: \partial C \rightarrow \mathbf{S}^{n-1}$ given by $f(x) = x/\|x\|$ is a homeomorphism.

PROOF. Since f is the composition of the inclusion $\partial C \hookrightarrow \mathbf{R}^n - \{0\}$ with the radial retraction $r: \mathbf{R}^n - \{0\} \rightarrow \mathbf{S}^{n-1}$, it is continuous. Proposition 16.2 implies that f is one-one, and f is obviously onto. By Theorem 7.8, f is a homeomorphism. $\square$

16.4. Theorem. A compact convex body C in $\mathbf{R}^n$ with nonempty interior is homeomorphic to the closed n -ball, and $\partial C \approx \mathbf{S}^{n-1}$.

PROOF. By translation, we can assume the origin is in the interior of C . Let $\mathbf{D}^n$ denote the unit disk in $\mathbf{R}^n$ and let f be as in Proposition 16.3. Then the function $k: \mathbf{D}^n \rightarrow C$ given by $k(x) = \|x\| f^{-1}(x/\|x\|)$ for $x \neq 0$ and $k(0) = 0$ clearly maps $\mathbf{D}^n$ onto C and is continuous everywhere except possibly at the

origin. However, since C is compact, there is a bound M for $\{\|x\| \mid x \in C\}$. Then $\|k(x)\| \leq M \cdot \|x\|$ which implies continuity at the origin. It is also clear that k is one-one, and hence it is a homeomorphism by Theorem 7.8. $\square$

17. The Baire Category Theorem

Often, one is interested in a condition on points of a space that is satisfied by an open dense set of points. For example, if $p(x_1, \dots, x_n)$ is a polynomial function on $\mathbf{R}^n$ then the condition $p(x) \neq 0$ has this property, and a special case of that is the determinant function on square matrices. If one has two such conditions then the set of points satisfying both conditions is still open and dense. The same, then, is true for any finite number of such conditions. But what of a countably infinite number of such conditions? Certainly, one cannot expect that the set of points satisfying all the conditions is open, but the density of this set does survive for a wide class of spaces, as we show in this section. This fact has many important consequences in analysis as well as in topology.

17.1. Theorem (Baire Category Theorem). *Let X be either a complete metric space or a locally compact Hausdorff space. Then the union of countably many nowhere dense subsets of X has empty interior.*

PROOF. Let U be an open subset of X and suppose that $A_i \subset X$ is nowhere dense ($i = 0, 1, \dots$). Construct a sequence of nonempty open sets $V_1, V_2, \dots$, such that $\bar{V}_{i+1} \subset V_i - \bar{A}_i$, where $V_0 = U$. (In the complete metric case, this can be achieved by taking $V_{i+1} = B_\epsilon(x)$ for some $x \in V_i - \bar{A}_i$ such that $B_{2\epsilon}(x) \subset V_i - \bar{A}_i$.)

If X is locally compact then also construct the V_i so that $\bar{V}_1$, and hence each $\bar{V}_i$, is compact. Then the $\bar{V}_i$ satisfy the finite intersection property and so $\emptyset \neq \bigcap \bar{V}_i \subset U - \bigcup \bar{A}_i$.

If X is complete metric, then also construct the V_i so that $\text{diam}(V_i) < 2^{-i}$. Then a sequence of points $x_i \in V_i$ is Cauchy since, for $i < j$, $\text{dist}(x_i, x_j) \leq \text{diam}(V_i) < 2^{-i}$. Then $x_n \in \bar{V}_i$ for all $n \geq i$ and so $x = \lim(x_n) \in \bar{V}_i$ for all i . Thus $x \in \bigcap \bar{V}_i \subset U - \bigcup \bar{A}_i$.

In both cases this shows that $U \not\subset \bigcup \bar{A}_i$. Since U is an arbitrary open set, we conclude that $\text{int}(\bigcup \bar{A}_i) = \emptyset$. $\square$

The word “category” in the theorem refers to the following definition:

17.2. Definition. A subset S of a space X is said to be of *first category* if it is the countable union of nowhere dense subsets. Otherwise it is said to be of *second category*. A set of second category is said to be *residual* if its complement is of first category.

Thus Theorem 17.1 can be rephrased: “An open subset of a complete metric space, or a locally compact Hausdorff space, is of second category in itself.”

It is also worth while to state the contrapositive of Theorem 17.1:

17.3. Corollary. *Let X be either a complete metric space or a locally compact Hausdorff space. Then the intersection of any countable family of dense open sets in X (i.e., a residual set) is dense.* $\square$

We close this section with some applications of this result. The first application deals with pointwise limits of functions.

17.4. Corollary. *If $\{f_n\}$ is a sequence of continuous functions $f_n: X \rightarrow Y$ from a complete metric space X to a metric space Y and if $f(x) = \lim f_n(x)$ exists for each x then the set of points of continuity of f is residual and hence dense.*

PROOF. For positive integers m, k let

$$U_{m,k} = \bigcup_{n \geq m} \{x \mid \text{dist}(f_n(x), f_m(x)) > 1/k\}$$

which is open. Since

$$\bigcap_{m \geq 1} U_{m,k}$$

consists of points where $f_n(x)$ does not converge, it is empty. It follows that

$$\bigcap_{m \geq 1} \bar{U}_{m,k} \subset \bigcup_{m \geq 1} (\bar{U}_{m,k} - U_{m,k})$$

which is a countable union of nowhere dense sets. Therefore

$$\bigcup_{k \geq 1} \bigcap_{m \geq 1} \bar{U}_{m,k}$$

is also a countable union of nowhere dense sets. Thus its complement

$$C = \bigcap_{k \geq 1} \bigcup_{m \geq 1} \text{int} \left(\bigcap_{n \geq m} \{x \mid \text{dist}(f_n(x), f_m(x)) \leq 1/k\} \right)$$

is residual. But $y \in C$ means that

$$\forall k \geq 1, \exists m \geq 1 \exists \delta > 0 \exists \text{dist}(x, y) < \delta \Rightarrow \forall n \geq m, \text{dist}(f_n(x), f_m(x)) \leq 1/k.$$

Hence, for such k, m, δ and $\text{dist}(x, y) < \delta$ we have that $\text{dist}(f(x), f_m(x)) \leq 1/k$ and also that $\text{dist}(f(y), f_m(y)) \leq 1/k$. By taking δ smaller, if necessary, we can also assure that $\text{dist}(f_m(x), f_m(y)) \leq 1/k$ by the continuity of f_m . Therefore $\text{dist}(f(x), f(y)) \leq 3/k$ for these choices, showing that f is continuous at y . (We hasten to point out that the set of points of discontinuity of f , while of first category, can well be dense. It is not hard to produce such examples.) $\square$

17.5. Corollary. *There exists a connected 2-manifold (i.e., a Hausdorff space in which each point has a neighborhood homeomorphic to the plane) with the following properties:*

- (1) *it has a countable dense set;*
- (2) *it has an uncountable discrete subset, and hence is not second countable; and*
- (3) *it is not normal, and hence not metrizable.*

PROOF. We will describe a similar manifold M “with boundary.” The desired manifold can then be obtained by “doubling” M ; i.e., taking two copies of M and identifying their boundaries.

First we describe the point set of M . There are two types of points. The first type consists of the points in the upper half space of the plane, i.e., $\{(x, y) | y > 0\}$. The second type of point is a ray (but we are describing a *single point* of M) from a point of the x -axis pointing into the upper half space.

To describe the topology on this set of points, we shall give a neighborhood basis at each point. For points in the upper half space, we use the usual topology of the plane. For a point corresponding to a ray r from a point x on the x -axis, we take, for a basic neighborhood, the set of points in the upper half plane in the “wedge” between two rays surrounding r and of distance (in the sense of the plane) less than ϵ from $(x, 0)$ together with the points of the second kind consisting of the rays from $(x, 0)$ and lying in the mentioned “wedge.” (See Figure I-7.)

To see that this really is a 2-manifold, consider the map ϕ from the upper half plane to itself given by $\phi(x, y) = (x/y, y)$. It is easy to verify that this is a homeomorphism on the upper half plane. Moreover, it maps rays from the origin to vertical lines. Thus the point of M corresponding to a ray from the origin can be thought of as the point on the x -axis attached to the vertical corresponding to the ray under ϕ . Under this correspondence it is evident that the topology becomes the ordinary topology of the closed half plane. This shows that a neighborhood of any point of M given by a ray from the origin, is indeed homeomorphic to an ordinary neighborhood of a boundary point in closed half space, a manifold with boundary. Rays from other points on the original x -axis can be treated similarly.

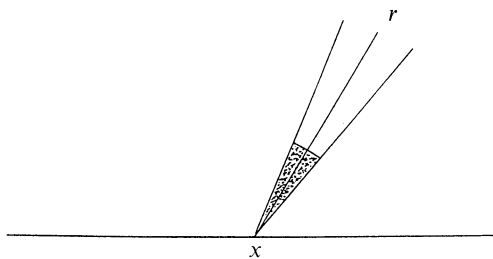


Figure I-7. Construction of a strange manifold.

We now verify the claims for this space. The points in the upper half plane with rational coordinates clearly give a countable dense set, proving (1).

Consider, for each x , the point of the second kind corresponding to a vertical ray from $(x, 0)$. Let S be the collection of these points. For any one of them a “wedge” neighborhood of that point intersects S in that point alone. Thus that point is itself an open subset of S . Thus S is discrete, and it is in one-one correspondence with the real axis, and so is uncountable. If M were second countable then any subspace, such as S , would also be second countable (just intersect the basis with the subspace), and that is not true of S .

Finally, we must show that M is not normal. In fact, let A be the subset of S , above, corresponding to rational x , and B that corresponding to irrational x . These are both closed subsets of M and are disjoint. We claim that it is impossible to separate them by disjoint open supersets. Suppose $U \subset M$ is an open set containing B and disjoint from A . For a point $x \in S$, and integer $n > 0$, let $W_n(x)$ be the wedge of angle π/n and radius $1/n$ about the vertical ray from x . Then define

$$T_n = \{x \in B \mid W_n(x) \subset U\}.$$

Then the sets T_n together with the singletons $\{x \in A\}$ comprise a countable collection of sets whose union is $A \cup B$, the real axis. Since the real line in its ordinary topology is complete metric, the closure in $\mathbf{R}$ of one of these sets must contain an interval. This is not true of the singletons, so it must be that $\bar{T}_n \supset (a, b)$ for some n and interval. But then for any “rational” $q \in (a, b) \cap S$ it is clear that every neighborhood in M of q must intersect U . $\square$

The manifold just described is, in fact, a differentiable manifold (see Chapter II) except for failure to be second countable. Indeed, it is the strange properties of examples like this that lead to the restriction to second countable spaces in the definition of a differentiable manifold.

Note that this manifold is a subspace of a normal space, its one-point compactification. Thus, a subspace of a normal space need not be normal.

17.6. Corollary. *In the space $\mathbf{R}^I$ of continuous functions $I \rightarrow \mathbf{R}$ in the uniform metric, the set of functions which are nowhere differentiable is dense. Indeed, it is residual in $\mathbf{R}^I$.*

PROOF. For a positive integer n , consider the set

$$U_n = \left\{ f \in \mathbf{R}^I \mid \forall t \in I, \exists s \neq t \text{ in } I \ni \left| \frac{f(t) - f(s)}{t - s} \right| > n \right\}.$$

We claim that this is open. To see this, note that for a given $f \in U_n$ and $t \in I$, there is an $\epsilon > 0$ and an $s \neq t$ such that

$$\left| \frac{f(t) - f(s)}{t - s} \right| > n + \epsilon.$$

Then, for some such $s = s(t)$ and $\epsilon = \epsilon(t)$, there is an open neighborhood V_t of t such that $s(t) \notin \bar{V}_t$ and such that

$$\left| \frac{f(t') - f(s)}{t' - s} \right| > n + \epsilon$$

for all $t' \in V_t$. The V_t cover I so that some finite union $V_{t_1} \cup \dots \cup V_{t_k} \supset I$. Let $\epsilon = \min \epsilon(t_i)$, $\delta = \min \text{dist}(s(t_i), \bar{V}_{t_i})$, and suppose that $\|f - g\| < \epsilon\delta/2$. Then, for any $t \in I$, we have $t \in V_{t_i}$ for some i and, for $s = s(t_i)$, we have

$$n + \epsilon < \left| \frac{f(t) - f(s)}{t - s} \right| \leq \left| \frac{f(t) - g(t)}{t - s} \right| + \left| \frac{g(t) - g(s)}{t - s} \right| + \left| \frac{g(s) - f(s)}{t - s} \right|.$$

Since $|t - s| \geq \delta$, the first and third terms on the right are each at most $(\epsilon\delta/2)(1/\delta) = \epsilon/2$. It follows that

$$\left| \frac{g(t) - g(s)}{t - s} \right| > n + \epsilon - \epsilon = n,$$

and hence that $g \in U_n$. Therefore, U_n is open as claimed.

Next we claim that each U_n is dense. To see this, let $f \in \mathbf{R}^I$ and $\epsilon > 0$ be given. Let m be so large that $2/m < \epsilon$. By uniform continuity of f there is a k so large that

$$|x - y| \leq 1/k \Rightarrow |f(x) - f(y)| \leq 1/m.$$

Also, take k so large that $k > nm$. Let $a_i = i/k$, $b_i = a_i + 1/(3k)$, $c_i = a_i + 2/(3k)$, and $y_i = f(a_i)$. Consider the interval $[a_i, a_{i+1}]$. Define a function g on this interval whose graph consists of the three line segments (a_i, y_i) to $(b_i, y_i - (1/m))$ to $(c_i, y_i + (1/m))$ to (a_{i+1}, y_{i+1}) ; see Figure I-8. These fit together to define g on all of I . By construction, $\|f - g\| \leq 2/m < \epsilon$. Let $t \in [a_i, a_{i+1}]$. If $g(t) > y_i$

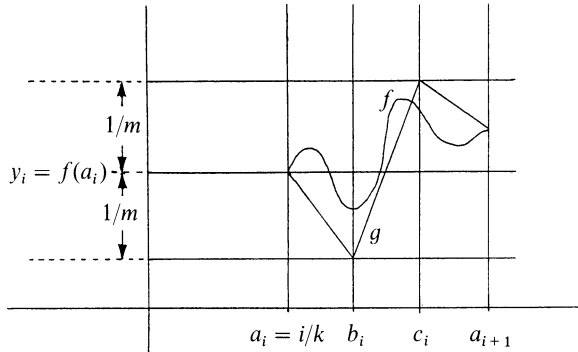


Figure I-8. Creating a nowhere differentiable function.

then take $s = b_i$. Otherwise, take $s = c_i$. Then

$$\left| \frac{g(t) - g(s)}{t - s} \right| \geq \frac{1/m}{1/k} = \frac{k}{m} > \frac{nm}{m} = n.$$

Hence $g \in U_n$ and $\|f - g\| < \epsilon$, concluding the proof that U_n is dense.

Since $\mathbf{R}^I$ is a complete metric space (prove it) we conclude that $A = \bigcap U_n$ is residual. We claim that any function $f \in A$ is nowhere differentiable. Suppose, on the contrary, that f is differentiable at some $t \in I$. Then $|(f(s) - f(t))/(s - t)|$ has a limit as $s \rightarrow t$ and so it is bounded for all $s \in I$, $s \neq t$. If n is larger than such a bound then it follows that $f \notin U_n$, a contradiction. $\square$

PROBLEMS

- Below is an outline of a more elegant proof of the fact that U_n is open in the proof of Corollary 17.6. Justify all statements made here.
 - The function $\mathbf{R}^I \times I \rightarrow \mathbf{R}$ taking $(f, t) \mapsto f(t)$ is continuous.
 - For $\Delta = \{(x, x) | x \in I\}$, the function $F: \mathbf{R}^I \times (I \times I - \Delta) \rightarrow \mathbf{R}$ taking $(f, s, t) \mapsto |(f(t) - f(s))/(t - s)|$ is continuous.
 - The map $\Phi: \mathbf{R}^I \times (I \times I - \Delta) \rightarrow \mathbf{R}^I \times (\mathbf{R} - \{0\})$ taking $(f, s, t) \mapsto (f, t - s)$ is closed.
 - The projection $p: \mathbf{R}^I \times (\mathbf{R} - \{0\}) \rightarrow \mathbf{R}^I$ is open.
 - $U_n = p(\mathbf{R}^I \times (\mathbf{R} - \{0\}) - \Phi F^{-1}[0, n])$ which is open.
- Let X be a complete metric space and let $\mathbf{R}^X$ be the set of continuous functions $X \rightarrow \mathbf{R}$. Let $S \subset \mathbf{R}^X$ be a collection of maps $f: X \rightarrow \mathbf{R}$ such that $\{f(x) | f \in S\}$ is bounded for each $x \in X$. Show that there is an open set $\emptyset \neq U \subset X$ and a number B such that $|f(x)| \leq B$ for all $x \in U$ and $f \in S$.
- An upper semicontinuous function of a real variable is a real valued function f on $\mathbf{R}$ such that $f^{-1}(-\infty, r)$ is open for all real r . If $f: \mathbf{R} \rightarrow \mathbf{R}$ is upper semicontinuous, show that there is some open interval (a, b) on which f is bounded below.
- Show that the set of points of continuity of an upper semicontinuous function is residual.